UNIVERSITI TEKNOLOGI MALAYSIA
**Department of Computer Science**
**Faculty of Computing**

# Project Task 6a

(Group Project)

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| **Programme** | : | Bachelor of Computer Science (*Data Engineering*) |
| **Subject Code** | : | SECR1213 |
| **Subject Name** | : | Network Communications |
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# Abstract

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# 2. Introduction

As the Faculty of Computing (FC) might anticipate a 15% growth in students and academic staff over the next four years, there is a need to establish an upgraded infrastructure that can accommodate this growth. To support this expansion, FC plans to construct a new two-storey building that will house multiple laboratories, a hybrid classroom, Video Conferencing Room, and a student lounge, all requiring a robust and scalable network infrastructure. The aim of this project is to develop a high-performance network that ensures seamless connectivity, strong security, and ease of management while maintaining cost-effectiveness. The network must also incorporate high-speed wired and wireless access, scalable network architecture, optimized network routes to support modern educational needs in alignment with the Fourth Industrial Revolution (4IR).

The scope of this project includes planning and designing the network infrastucture, structuring network architecture, integrating advanced technology into the labs and rooms, budgeting, implementing the network and calculating network routes to ensure optimal performance. The objectives of our team are to establish efficient communication across all facilities, enhance network security, and ensure scalability for future expansion. By leveraging both our knowledge and advancements in networking technologies, we aim to create an innovative, connected educational environment that not only fulfils the current demands of FC but also positions it for long-term technological advancements and growth.

# 3. Project Background and an overview of the client's status and issue

The Faculty of Computing (FC) is currently facing a significant challenge as FC is projected to have 15% increase in students and staff population over the next four years. To accommodate this expansion, FC plans to construct a new two-storey building that will house 2 general purpose labs, a Cisco Network lab and an Embedded lab, a video conferencing room, a student lounge and hybrid classroom. Each lab and the student lounge will be 14mx10m in size and all interconnected via Wi-Fi. However, the successful implementation of this new facility depends on a well-designed, high-performance network infrastructure that can support the faculty's evolving technological and educational needs. As education moves towards digital learning and IoT-integrated environments, FC requires a scalable, secure, and efficient network system that enhances connectivity while being cost-effective.

Currently, FC need a better equipment that will allow for this growth. All stakeholders want to prepare for future needs to the extent possible at the best value possible which particularly ensure a reliable, efficient and security vulnerable network infrastructure that hinder its ability to support modern digital learning tools. We have to make sure the network is easy to manage and scalable, capable of enhancing overall performance, and equipped with robust security measures to protect against cyber threats, including internet worms, denial-of-service attacks, and e-business application vulnerabilities. Additionally, the network must provide high-performance support to the core backbone, ensuring uninterrupted access to critical resources such as virtual learning platforms, hybrid applications, and workshops. The transition from the current infrastructure to the upgraded system must be seamless, ensuring that users continue to have uninterrupted access throughout the implementation phase.

In short, the development of this new 2-storey building, featuring different labs with a strong focus on network infrastructure is designed to support the evolving needs of FC's academic community while maintaining a balance between innovation, long-term scalability, and cost efficiency for the future.

# 4. Compiled Solution

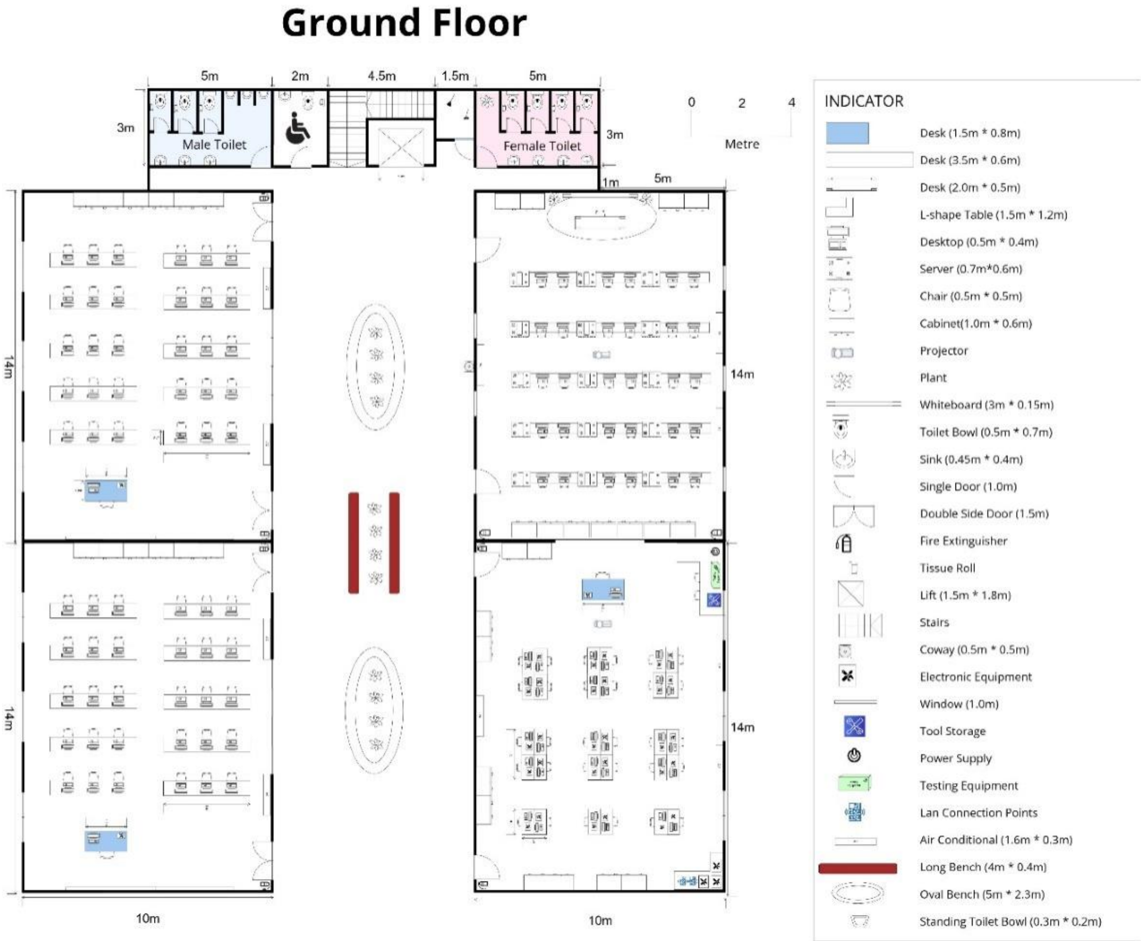
## 4.1 Task 1

### 4.1.1 Brief Explanation for The Task

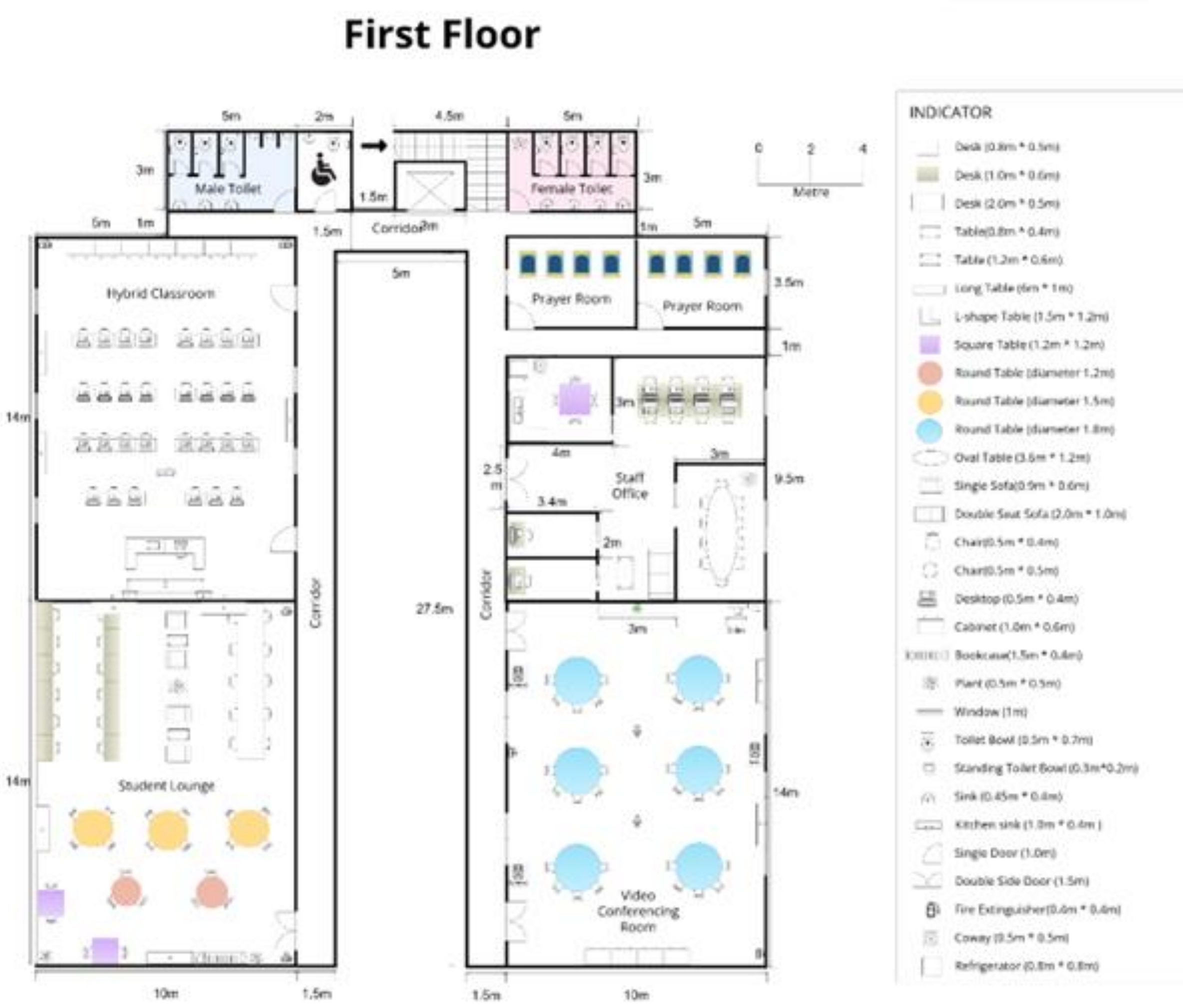
Task 1 which is project setup focuses on creating a team and establishing the foundation setup for a small network design project. This project is to accommodate the growth of the students over the next 20 years, as the computing field grows and moves forward into the future. We formed our group of four students, including Chau Ying Jia, Poh Lok Yee, Sabrina Heng Wei Qi and Woo Cheng Shuan. We selected 4Fabs as our group name. The team's objective is to understand the case study and design a new 2-storey building. This building must house 4 labs which is 2 general purpose labs, 1 Cisco Network lab and 1 Embedded lab. Besides, it must also include 1 video conferencing room, 1 hybrid classroom and a student lounge. Additionally, our group members will document all group meetings in detailed minutes, outlining discussions, decisions and responsibilities.

Group members have to research and based on each room different purpose, design in detail including what equipment and the fair positioning of them in the rooms. The floorplan is also part of our task to design a building layout based on the required room with additional for the daily purposes of students and lecturer to meet the needs of daily users. This project focuses on planning and designing an infrastructure that supports the faculty's growth objectives and technological requirements, ensuring it remains both efficient and secure. The success and grade of the project will depend on the team's ability to collaborate effectively and produce a well-thought-out, high-quality design.

# 4.1.2 Floor plan for Ground Floor



### 4.1.3 Floor plan for First Floor



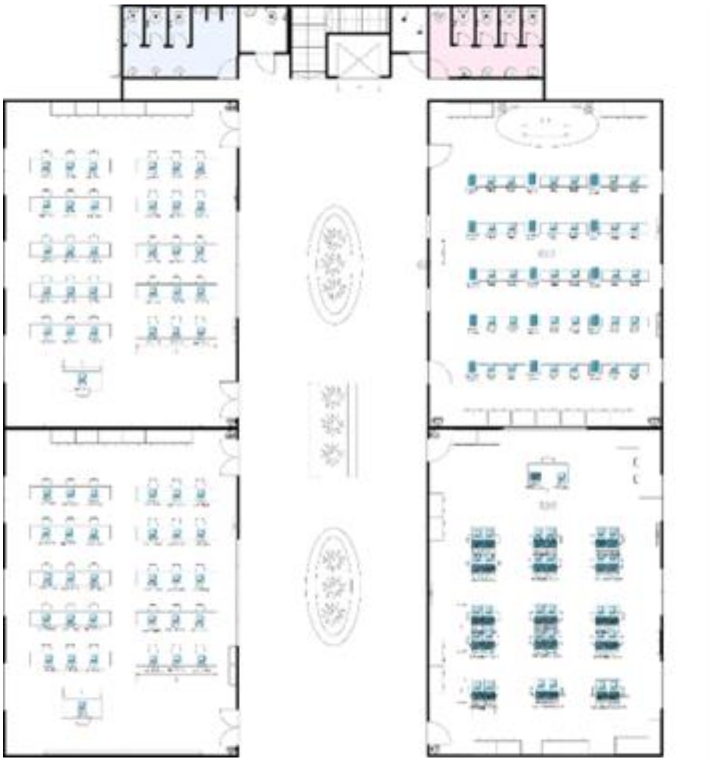
### 4.1.4 Brief Explanation for The Layout

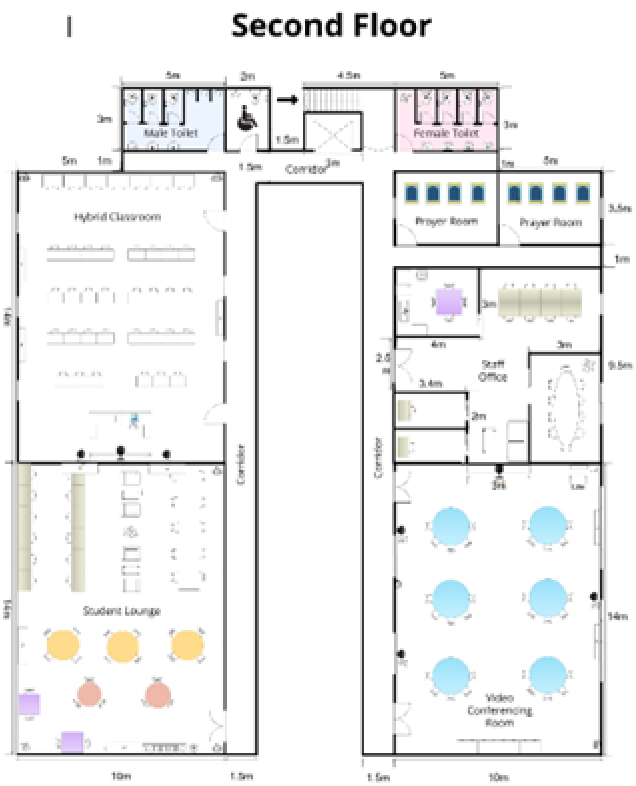
We are establishing four laboratories on the ground floor which are two general laboratories, a Cisco Network Lab, and an Embedded Lab. The two general labs are positioned on the left side of the building, while the Cisco Network Lab and the Embedded Lab are located on the right side. Centrally situated is a lift, surrounded by stairs. At the back of the left wing, there is a male restroom, along with an accessible restroom next to the stairs. Conversely, the female restroom is located on the opposite side of the stairs. The center of the building features greenery and benches for students to utilize.

For the first floor, we have designated spaces for the Student Lounge, Staff Office, Hybrid Classroom, and Video Conferencing Room. The Student Lounge is strategically positioned on this level due to its quieter environment, minimizing people passing by and creating a more conducive atmosphere for students. As illustrated, the Student Lounge and Hybrid Classroom are located on the left side, while the Staff Office and Video Conferencing Room are situated on the right side of the building. Additionally, we have allocated two prayer rooms beside the staff room for the male and female Muslims separately. Finally, the design and placement of both the female and male restrooms mirror those on the ground floor.

### 4.1.5 Enhanced Floorplan by using Packet Tracer

Ground floor





### 4.1.6 Task 1 Reflection

This project was an insightful and valuable learning experience, as it required our team to collaborate, research and apply our technical knowledge to design a two-story building with an optimized network infrastructure. The process of planning and implementing the floor plan using tools

## 4.2 Task 2

### 4.2.1 Question

1. **What are the current pain points with the existing networking setup?**

   There are frequent connectivity issues during peak usage times, especially in high-density areas like lecture halls and labs and also some areas with lower or even no connection. We also experience slower speeds and intermittent dropouts during video conferences and virtual meetings, which can disrupt communication and collaboration.

2. **What types of network devices will be required for the Cisco Network lab, video conferencing room and embedded lab?**

   The Cisco Network Lab will require appropriate networking equipment, such as routers, switches, and firewalls, to facilitate the teaching of Cisco-related concepts.[6] Similarly, the video conferencing room will necessitate high-quality audiovisual components, including cameras, microphones, speakers, and a reliable, high-bandwidth internet connection, to enable effective remote communication and collaboration. Lastly, the embedded lab will need a range of specialized hardware, including logic analyzer and Raspberry Pi kits, microcontrollers, digital oscilloscopes, and sensor modules, in addition to relevant software tools like MATLAB, LabVIEW, and IoT cloud platforms, to support the hands-on exploration and development of embedded systems.[7, 8]

3. **What are the essential network and hardware requirements for an effective and high quality video conferencing room?**

   The hardware requirement for an effective and high-quality video conferencing room is having a high-resolution webcam. It is essential for clear video communication, as it captures subtle details like facial expressions and body language, enhancing the authenticity of virtual interactions. The sufficient network bandwidth is also critical to achieve the requirement. For example, a moderate-quality 720p video at 15 frames per second requires about 1 Mbps, while a high-quality 4K video at 30 frames per second demands approximately 4 Mbps.The amount of internet strength needed depends on how clear you want the video to be. Additionally, when multiple participants join a video call, higher bandwidth is necessary to maintain smooth connectivity. Even with strong Wi-Fi coverage, simultaneous usage can strain the network, the network must be capable of handling concurrent usage to maintain uninterrupted and high-quality video conferencing.

4. **What network services will be offered to support teaching and learning activities?**
Provide virtual learning platforms hosted on local servers or the cloud, secure access to digital libraries, research databases, and online courses. Implement streaming services for live lectures and virtual labs to enhance interactive learning. For example, Cisco Education which securely connects educators, students, and administrators with transformative technologies that power inclusive learning for all[15].

5. **How can the building's wireless network support high-density areas like the student lounge and labs?**

To effectively support high-density areas, such as the student lounge and labs, the building's wireless network should utilize enterprise-grade access points equipped with multi-user, multiple-input, multiple-output technology and band steering capabilities. Strategically positioning the access points can optimize wireless coverage. Additionally, implementing Wi-Fi 6 access points can facilitate simultaneous connections for numerous devices in each floor.[10] Further, segregating the wireless network into separate service set identifiers for students, staff, and guests, while imposing bandwidth restrictions for non-critical users, can help manage network usage efficiently.

6. **What type of network architecture (e.g., star, mesh, bus, or hybrid) would best suit the faculty's needs, considering factors like scalability, reliability, performance, budget, and ease of maintenance?**

A hybrid network architecture that combines star and mesh topology is the most suitable design for the new faculty. On the first floor, star topology will be used to connect all devices such as printers, desktops and wireless access points to a central switch to ensure the effectiveness and localized connectivity. Otherwise, mesh topology will be used to link Floor 2's switches to Floor 1's main switch using a high-speed backbone such as fiber-optic or high-capacity Ethernet.This design balances scalability, performance, and cost, enabling easy additions of devices and robust communication between floors.

7. **What is the best method of connecting devices to the network (e.g., wired, wireless, or a combination of both)?**

The best method for connecting devices to the network is a combination of both wired and wireless connection depending on the type of device and usage. For wired networks, devices are physically connected by using cables to a network switch or router.[20] Wired networks are suitable for devices such as printers, desktop as they are able to provide a stable and high-speed connection with low latency. However, wireless connection using electromagnetic waves to transmit data from end-to-end. Devices like laptops, tablets and smartphones are more suitable to use wireless connection as it is more flexible without using physical cable and ease of installation[20]. Otherwise, Wireless Access Points (APs) are strategically placed throughout the building to provide seamless coverage, allowing users to connect to the network with more stability.

8. **What kind of wireless network setup is required to provide seamless connectivity across the building?**

   To provide seamless wireless connectivity across the faculty building, deploy multiple access points (APs) to ensure consistent coverage in all areas. Use wireless repeaters or extenders to eliminate dead zones in distant or obstructed spaces. Incorporate network switches and controllers for centralized management of APs, streamlining configuration, and improving network performance. Additionally, consider using directional or omnidirectional antennas based on the coverage requirements and ensure all devices are equipped with network interface cards (NICs) for compatibility with the wireless network[12].

9. **What type of network cable is the most appropriate for connecting the computers, and what are the reasons for choosing it?**

   Cat6 Ethernet cables are the most appropriate type of cabling for the network setup. Cat6 cables support speeds up to 1Gbps, which is sufficient for typical educational applications and ideal for labs with the need of high speed connectivity[16]. Additionally, Cat6 cables have a good balance between performance and cost as they are widely available and easy to source. Cat6 cables are compatible with existing networking equipment such as routers and switches that support Gigabit Ethernet, ensuring that the faculty’s network instructure work efficiently without requiring an upgrade[16].

10. **How can network efficiency and low latency be achieved?**

    Low latency and network efficiency require the use of several techniques. A network design that is well-structured, like a core-distribution-access topology, guarantees effective traffic flow by lowering congestion and the distance that data must travel. Important applications like video conferencing may function without interruption thanks to network segmentation using VLANs, which helps isolate various traffic types.[2] By prioritising time-sensitive traffic, Quality of Service (QoS) implementation guarantees minimal latency for applications with high demand.[1] To maximise speed and reduce delays, high-performance gear is necessary. Examples of this hardware include fiber-optic cabling for inter-floor connections, low-latency access points, and gigabit or 10-gigabit Ethernet switches. The network is further optimised by traffic prioritization including load balancing, traffic shaping, and redundancy, which distribute traffic uniformly and offer backup routes to avoid congestion.[1]

**11. What bandwidth and internet connectivity are required to support the faculty's operations?**

A minimum bandwidth of 1 Gbps is recommended, with provisions to scale to 10 Gbps in the future[11]. Bandwidth is critical to the success of hardware initiatives that attempt to provide ubiquitous access to learning, such as Bring Your Own Device (BYOD) and 1:1 laptop and tablet programs. The amount of bandwidth needed is obviously driven by the types of networked activities in which the user is engaged such as downloading content, online professional developments and online testing.

**12. How can we ensure that the wireless network covers the entire faculty building effectively?**

To ensure that the coverage wireless network is able to support the area of the faculty, a thorough evaluation of signal strength across all areas, including floors, common spaces and specific rooms is essential[19]. This evaluation involves determining the quantity of current Access Points (APs) and their effective range to establish the extent of network coverage [19]. Detect and scrutinize dead zones, areas with weak or non-existent signal strength [19], to pinpoint areas in need of improved coverage. Considering the characteristics of the wireless network, including factors like Wi-Fi channels, network authentication, and security protocols, make necessary adjustments to optimize coverage and ensure a dependable connection. Based on the findings and analysis, identify suitable new locations for the placement of APs to enhance coverage and address the previously identified dead zones. Carefully position the new APs to ensure high-quality speed and coverage precisely where it's needed.

**13. How should the building's network be secured against external and internal threats?**

The Cybersecurity and Infrastructure Security Agency (CISA) encourages users and network administrators to implement the segment and segregate networks and functions to better secure their network infrastructure. Proper network segmentation is an effective security mechanism to prevent an intruder from propagating exploits or laterally moving around an internal network. A securely segregated network can contain malicious occurrences, reducing the impact from intruders in the event that they have gained a foothold somewhere inside the network[13]. Moreover, implementing a

Virtual Private Network (VPN) is essential[14]. A VPN encrypts data transmissions and masks users' IP addresses, providing secure access to the network, especially for remote users or those connecting via public Wi-Fi. This helps protect sensitive information from unauthorized access and cyber threats.

14. **What disaster recovery plan is needed for network downtime?**

    Planning for disaster recovery is essential to ensure an organization can maintain operations during unexpected events such as natural disasters, cyberattacks, or other emergencies. Regular data backups can ensure the important files and settings are safe and can be recovered easily. Implementing a backup network system, like additional routers and internet connections can keep the network running if the main system fails[17]. The real-time monitoring tools play a crucial role in detecting issues early by sending alerts, so issues can be addressed quickly[17]. Futhermore, Uninterruptible Power Supplies(UPS) can be used to protect devices from power outages as it can give more time to fix the issue without losing connectivity[18]. These measures collectively strengthen the organization's resilience and minimize the impact of disruptions.

15. **What monitoring tools are recommended for detecting and managing network issues?**

    Several monitoring technologies can be used to provide real-time insights, identify bottlenecks, and guarantee optimal performance in order to detect and address network issues efficiently. With comprehensive data on bandwidth utilisation, latency, and device condition, SolarWinds Network Performance Monitor is a well-liked tool for keeping an eye on the functionality and health of network devices.It can detect and solve network issues before they cause downtime.[3] With customisable alerts to inform administrators of any problems, PRTG Network Monitor provides a complete solution for tracking network traffic, server performance, and device health. Another open-source tool for network and infrastructure monitoring is Nagios, which can measure a number of metrics like response times, availability, and uptime.[3] In order to solve problems like network congestion or security vulnerabilities, network administrators can collect and examine packets using Wireshark, which allow us to capture and analyze network traffic. [4]

### 4.2.2 Feasibility Assessment

Feasibility assessments are critical in determining whether a project is realistic and worth pursuing. After the Question and answer part, we can now evaluate and discuss the feasibility of our project. These are the four feasibility we are going to determine and evaluate:

1. Technical Feasibility

   Based on the information gathered, the project appears to be technically feasible. The proposed network architecture features a fiber-optic backbone and Cat6 cabling to support high-speed connectivity and scalability for faculty. We also select enterprise-grade access points with Wi-Fi 6 to ensure coverage in high-density areas like labs and lounges to meet future demands. The integration of devices such as Raspberry Pis, microcontrollers, and Cisco equipment meets the labs' technical and educational needs. Lastly, the adoption of modern technologies like VLANs for traffic segmentation, Quality of Service (QoS) for latency-sensitive applications, and monitoring tools like SolarWinds and Nagios enhance network reliability and performance.

2. Economic Feasibility

   It is essential to evaluate the costs and benefits of the proposed investment to assess the economic feasibility of the project. The initial costs include purchasing equipment, software installation and software license fees while operational costs include maintenance, IT management staffing and power consumption. However, the project offers significant economic advantages, such as reduced downtime, increase productivity, and long-term savings from scalable and future-proof technology like Wi-Fi 6 and Cat6 Ethernet cabling. A cost-benefit analysis which compare the costs with the benefits it will provide will helps in ensuring the project stays within budget while achieving the faculty's goals effectively.

3. Operational Feasibility

   The operational feasibility of the project appears promising. By using a hybrid network architecture which combines star and mesh topologies helps to ensure the connectivity and seamless communication between floors. Otherwise, high-performance hardware such as managed switches, routers and Wi-Fi 6 access points support efficient wired and wireless connections. Moreover, for solving traffic segregation, we have included VLANs and Quality of Service(QoS) to keep the lower latency while using video conferencing and hybrid labs. Monitoring tools like SolarWinds enable proactive issue management.While strong security standards protect data,but disaster recovery methods, such as backups and redundant systems can improve resilience.

4. Security Consideration

   To secure the faculty building's network, segmentation and segregation of networks are essential to contain threats and limit intruder movement within the network. Implementing Virtual Private Networks (VPNs) ensures encrypted communication for remote access and public wi-Fi usage. A robust disaster recovery plan is necessary such as redundant systems and regular backups, making the network secure and reliable for educational operations.

### 4.2.3 Task 2 Reflection

## 4.3 Task 3

### 4.3.1 Table of Devices Needed

| Devices | Description | Price per Unit | Quantity | Total price (RM) |
|---|---|---|---|---|
| Routers  | Ubiquiti EdgeRouter 12<br>– **System memory:**<br>– 1 GB DDR3 RAM<br>– **On-board flash storage:** 4 GB eMMC, 8 MB SPI NOR<br>– **Networking interface:** (10) GbE RJ45 ports, (2) 1 Gbps SFP ports<br>– **Management interface:** (1) RJ45 serial port<br>– **Power method:** External AC/DC power adapter, 24W, 24V, 1A (Included) , 24V Passive PoE<br>– **Power supply:** External AC/DC adapter<br>– **Passive PoE**<br><br>**Voltage range**<br><br>**Max. wattage per port:** DC/PoE passthrough<br><br>20W (24V)<br><br>– **Supported voltage range:** 9-30V DC<br>– Max. power consumption: 20W (Excluding PoE output) | 1,269 | 4 | 5076 |
| Switches  | S3400-24T4FP<br>- **Ports:** 24x 10/100/1000BASE-T RJ45 \| 4x 1G RJ45/SFP Combo<br>- **Fans:** 3 Built-in Fans<br>- **PoE Budget:** 370W<br>- **Material:** Metal<br>- **Features:**<br>1) Virtual deployment of dual switches offers flecible and highr-performance | 2033 | 13 | 26429 |

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|---|---|---|---|---|
| | network for serving all scenarios on campus<br>2) Support web management, firmware upgrade and configuration file upload/download via TFTP/HTTP/FTP server, and firmware auto upgrade Remote Monitoring (RMON). | | | |
| Patch Panels<br> | BELNET CAT6 24-Port Patch Panel<br>- **Type:** CAT6 24-Port Patch Panel<br>- **Form Factor:** 1U rack<br>- **Network Support:** Gigabit Network (Transmission up to 250MHz, supports speeds of 1G – 10G Mbps)<br>- **Performance Standard:** Tested according to ISO/IEC 11801 CLASS E PL2.<br>- **Front Panel:** Include label field for easy identification and have RJ45 front sockets with a smart shutter<br>- **Rear Side:** Equipped with a cable management metal bar for organizing and securing cables<br>- **IDC Termination:** Provide Termination Caps, compatible with 568A & 568B color codes, supports cable wire diameters from AWG23 to AWG26<br>- **Package includes** Screws set (x4), Cable ties (x24)<br>- **Weight:** 0.9kg<br>- **Dimensions:** 52x9.5x6cm | 378 | 13 | 4914 |
| Server<br> | Dell PowerEdge R750 Rack Server<br>- **Form factor**: 2U rack<br>- **Processors**: Up to 2 x 3rd Generation Intel® Xeon® Scalable processors; up to 40 cores per processor<br>- **Memory**: Up to 3200MT/s<br>- **Storage**: Up to 12 x 3.5-inch SAS/SATA (HDD/SSD), max 192TB | 30,895 | 4 | 123,580 |

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|---|---|---|---|---|
| | - **Networking**: 2 x 1 GbE LOM and 1 x OCP 3.0 (x8 PCIe lanes) (optional)<br>- **NVMe Support**: Up to 24x NVMe (HDD/SSD)<br>- **Management**: Embedded iDRAC9 | | | |
| Wifi Access Point<br> | Cisco Catalyst 9130AX series<br>- **Wireless Communication Standard:** 802.11ax<br>- **Antenna Type:** Internal antenna<br>- **Radio Chains and Streams:** 4x4 4MIMO<br>- **Frequency Band Class:** Dual-Band<br>- **Frequency:** 2.4GHz and 5GHz<br>- **Speed:** 5.38 Gbit/s<br>- **Bluetooth Standard:** Bluetooth 5<br>- **Client Capacity:** Supports over 1000 clients<br>- **Security:** WPA3, Cisco TrustSec and identity-based access | 3300 | 10 | 33000 |
| Projector<br> | Epson PowerLite X49<br>- **Brightness:** 3600 lumens<br>- **Connectivity:** HDMI,USB,VGA,Wireless<br>- **Resolution:** XGA (1024*768)<br>- **Contrast:** 22000:1<br>- **Lamp Life:** Up to 15000 hours<br>- **Audio:** Built-in 2W speaker<br>- **Projected distance:** 1.189-13.106 m<br>- **Features:** Technology: Digital Light Processing (DLP), sharper images, compact size | 1799 | 5 | 8995 |
| Desktop<br> | HP Elite Tower 800 G9 Desktop<br>- **Processor:** Intel® Core™ i5-13500<br>- **Graphics**: Intel UHD Graphics 770<br>- **Ports:** Provides a SuperSpeed USB Type-C port with 20Gbps | 4652 | 164 | 762928 |

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|---|---|---|---|---|
| | - **Memory and Storage:** 512GB SSD | | | |
| SmartBoard<br> | Huawei IdeaHub S2<br>- **Operating System:** HarmonyOS<br>- **Connectivity:** WiFi-6, Bluetooth, USB, HDMI<br>- **RAM:** 8GB<br>- **Storage:** 128GB<br>- **Resolution:** 4K Ultra HD (3840*2160)\<br>- **Display Size:** 65", 75", 86"<br>- **Camera:** 4K Video Camera<br>- **AI features:** Voice tracking, auto- framing and acoustic baffle<br>- **Latency:** Ultra-low 16ms | 15000 | 1 | 15000 |
| Uninterrupted Power supply (UPS)<br> | CyberPower UPS OL10KRT<br>– **Capacity:** 10kW<br>– **Topology:** Double-Conversion<br>– **Runtime:** expandable with additional battery packs<br>– **Remote Management:** PowerPanel software enables to monitor and mange UPS remotely<br>– **Ports:** USB, Serial, Relay, EPO<br>– **Energy saving:** ECO mode efficiency up to 97%<br>– **Fast Charge Technology:** Yes | 36000 | 1 | 36000 |
| Cat6 Ethernet Cable<br> | Mediabridge Ethernet Patch Cable (Cat6)<br>– **Brand:** Mediabridge<br>– **Connector Type**: RJ45<br>– **Cable Type:** Ethernet<br>– **Compatible devices:** computer, router, modem, television<br>– **Special feature:** high speed, data transfer, low voltage<br>– **High Speed:** Ultra Fast Throughput of 10 Gigabit per Second at 500 MHz<br>– **Use:** Easily handles the most demanding home use such as Gaming, High- | 80/ 15m | 1500m | 8000 |

| | | | | |
|---|---|---|---|---|
| | Definition Video Streaming, Cloud Computing etc.<br>– **Server applications:** 10 gigabit throughput at up to 250 MHz guarantees high-speed data transfer for server applications. Suitable for 10BASE-T, 100BASE-TX (Fast Ethernet), 1000BASE-T/1000BASE-TX (Gigabit Ethernet) and 10GBASE-T (10-Gigabit Ethernet) applications. | | | |
| Connector<br> | Cat6 RJ45 Modular Plug (8P8C)<br>- **Ethernet Type:** Rj45 Connector<br>- **LAN category**: Cat6, Cat6a<br>- **Contact Plating**: Gold<br>- **Mounting Type**: Cable Mount | 0.38 | 500 | 190 |
| Fiber Optic Cables<br> | SC APC Patch Cable, OS2<br>- **Fiber count:** 1 fiber<br>- **Wavelength**: 1310/1550nm<br>- **Insertion Loss:** <=0.3dB<br>- **Return Loss:** >=60dB | 14/ 1m | 50 | 700 |
| | | | TOTAL | 1024812 |

### 4.3.2 Major differences between the same devices from different brands

**Routers**

Here's a detailed comparison table between the **Ubiquiti EdgeRouter 12** and the **TP-Link Omada ER7206** routers, which are suitable for university faculty use.

| **Feature** | **Ubiquiti EdgeRouter 12** | **TP-Link Omada ER7206** |
|---|---|---|
| **Price** | MYR 2,200 - MYR 2,800 | MYR 1,500 - MYR 2,000 |
| **Ports** | 12 Gigabit Ethernet ports (1 SFP port for fiber) | 6 Gigabit Ethernet ports |
| **Processor** | 1.7 GHz Dual-Core Processor | 1.7 GHz Quad-Core Processor |
| **Wi-Fi Support** | No built-in Wi-Fi support, requires external APs | No built-in Wi-Fi support, requires external APs |

| | | |
|---|---|---|
| **Routing Protocols** | Supports static routes, OSPF, BGP, and more | Supports static routing, OSPF, RIP, and more |
| **Management** | Web-based interface (Unifi Controller can be used separately) | Centralized management via Omada SDN Controller (Cloud-based) |
| **VLAN Support** | Yes, supports advanced VLAN configurations | Yes, supports VLAN configuration with dynamic VLAN tagging |
| **Firewall** | Advanced firewall features, with DPI (Deep Packet Inspection) | Advanced firewall features, including DoS protection |
| **Security Features** | IP Filtering, DoS Protection, Stateful Packet Inspection | IP Filtering, DoS Protection, VPN passthrough, SPI |
| **VPN Support** | Supports OpenVPN, L2TP/IPsec, PPTP, and more | Supports OpenVPN, L2TP, PPTP, and IPsec VPN |
| **Power over Ethernet (PoE)** | Yes, on some ports (PoE+ on 2 ports) | Yes, PoE+ on 4 ports |
| **Performance (Routing Throughput)** | Up to 3.4 Gbps routing throughput | Up to 1.7 Gbps routing throughput |
| **Suitable for** | Larger setups with **high network traffic needs**, including more advanced routing | SMBs and faculties with moderate network traffic demands |

According to the comparison above, since our faculty plan to establish multiple labs, including a Cisco networking lab and a video conferencing room, indicates the need for large and complex setups. We require routers with additional ports and greater routing throughput, making the Ubiquiti EdgeRouter 12 an excellent option. It features six more Gigabit Ethernet ports than the TP-Link Omada ER7206 and provides a higher routing throughput of 3.4 Gbps, making it ideal for more extensive network configurations. Additionally, it offers advanced routing capabilities like OSPF and BGP, which are perfect for intricate network environments, along with limited PoE support on specific ports for external access points.

**Switches**

We have chosen **S3400-24T4FP** switches to use in this project. The major difference between S3400-24T4FP with other different brands we can compare is **NetGear ProSAFE GS728TP.**

| **Specification** | **S3400-24T4FP** | **NetGear ProSAFE GS728TP** |
|---|---|---|
| **Packet Buffer Size** | 500KB | 512KB |
| **Ports** | 24 x 1Gbe + 4 Combo Uplinks | 24 x 1Gbe + 4 Combo Uplinks |
| **Cooling** | 3 Built-in Fans | Fanless |
| **PoE Budget** | 370W | 380W |

| | | |
|---|---|---|
| **Power Consumption** | 400W | 300W |
| **Ability to Handle Traffic** | Yes with advanced L2+ Features | Yes with advanced L2+ Features |
| **Power Efficiency** | High | High |
| **Material** | Metals | Metal |
| **Price** | RM 2033 | RM 2900 |

Based on the comparison above, both are robust options for network environments, but the S3400-24T4FP offers a better balance of performance and affordability. Although the NetGear model has a slightly higher PoE budget of 380W and lower power consumption of 300W compared to S3400's 370W and 400W, the S3400 have 3 built-in fans for effective cooling, ensuring reliable operation under heavy loads, whereas the fanless design of NetGear may lead to potential overheating in high-demand scenarios. Both switches feature same ports number and advanced layer 2+ traffic management capabilities, but the S3400 achieves this at much lower price of RM2033, significantly undercutting the NetGear's RM2900. This makes the S3400-24T4FP the more cost-effective and practical choice for most network setup.

**Patch Panels**

For patch panels in this project, we have chosen BELNET CAT6 24-Port Patch Panel. The major difference between **BELNET CAT6 24-Port Patch Panel** with other different brands we can compare is **Mini-Com® Patch Panel by Panduit.**

| Specification | BELNET CAT6 24-Port Patch Panel | Mini-Com® Patch Panel by Panduit |
|---|---|---|
| **Category** | CAT6 | CAT6 |
| **Ports** | 24 | 24 |
| **Size (Rack Units)** | 1U | 1U |
| **Design** | Simple, functional front with label fields for port IDs. | Modular design with label fields for easy identification. |
| **Cable Management** | Includes a metal bar for cable management at the rear. | Modular design allows flexibility with cable routing. |
| **Shutter Protection** | Dust-protective shutters on RJ45 sockets. | No shutters |
| **Color-Coded Labels** | 568A & 568B for installation ease. | Provides clear labelling options for modular configurations. |
| **Wire Support** | Accepts AWG23 to AWG26 cables, solid or standard | Compatible with various cable types based on modular jacks |
| **Tool Compatibility** | Suitable for Krone and 110 punch-down tools | Compatible with modular components |
| **Construction Material** | Metal chassis for durability | Robust design with high-quality materials |
| **Testing** | Fluke-tested for performance assurance | Meets or exceeds industry standards like TIA/EIA |
| **Price** | RM 378 | RM 627.13 |

The major differences between devices from different brands lie in build quality, features, performance, cost and support. For our faculty, BELNET is more suitable as it focusses on affordability and basic functionality, suitable for smaller-scale setups. Although Panduit offer advanced functionality, better materials and robust support, but its price is much higher and will exceed our budget. In conclusion, BELNET will be our choice as it reaches our requirement and budget.

**Server**

For the servers used in this project, we have chosen **Dell PowerEdge R750 Rack Server**. The major difference between **Dell PowerEdge R750 Rack Server** with other different brands we can compare is **Lenovo ThinkSystem SR650**.

| **Specification** | **Dell PowerEdge R750 Rack Server** | **Lenovo ThinkSystem SR650** |
|---|---|---|
| **Form factor** | 2U rack | 2U rack |
| **Processors** | Up to 2 x 3rd Generation Intel® Xeon® Scalable processors; up to 40 cores per processor | Up to 2x 3rd Gen Intel® Xeon® Scalable processors |
| **Memory** | Up to 3200 MT/s | Up to 32x 128GB TruDDR4 3200MHz memory |
| **Storage** | Up to 12 x 3.5-inch SAS/SATA (HDD/SSD) max 192 TB | Up to 32x NVMe drives |
| **Networking** | 2 x 1 GbE LOM and 1 x OCP 3.0 (x8 PCIe lanes) (optional) | LOM adapter installed in the OCP 3.0 slot; PCIe adapters |
| **NVMe support** | Up to 24x NVMe (HDD/SSD) | Up to 32x NVMe drives |
| **Management** | Embedded iDRAC9 | Lenovo XClarity Controller |
| **Price** | RM30,895 | RM16,945 |

Based on the table above, the Dell PowerEdge R750 Rack Server is the more suitable option for the faculty due to its superior performance and advanced capabilities. With support for up to 40 cores per processor and versatile storage options, including SAS/SATA and NVMe drives, it is ideal for resource- intensive tasks such as simulation, virtual labs and research. The integrated iDRAC9 management system ensures efficient remote monitoring and troubleshooting, enhancing operational reliability. Additionally, its flexible networking options via PCle lanes provide scalable connectivity solution, making the Dell PowerEdge R750 a powerful and future-ready choice for a demanding academic environment. However, it comes at a higher cost compared to its counterpart.

**Wi-Fi Access Point**

For the Wi-Fi Access Point used in this project, we have chosen Cisco Catalyst 9130AX Series. The major difference between **Cisco Catalyst 9130AX Series** with other different brands we can compare is **Aruba 515**.

| **Specification** | **Cisco Catalyst 9130AX Series** | **Aruba 515** |
|---|---|---|
| **Wi-Fi Standard** | Wi-Fi 6 (802.11ax) | Wi-Fi 6 (802.11ax) |
| **Maximum Data Rate** | Up to 5.38 Gbps | Up to 4.8Gbps |

| | | |
|---|---|---|
| **Radio Chains and Streams** | 4x4:4 MU-MIMO | 4x4:4 MU-MIMO |
| **Antenna Type** | Internal and external antenna options available | Internal antennas |
| **Frequency Bands** | 2.4GHz and 5GHz Dual band | 2.4GHz and 5GHz Dual band |
| **Power over Ethernet (PoE)** | PoE+, UPOE and UPOE+ support | Supports PoE (802.3af/802.3at) |
| **Client Capacity** | Supports over 1000 clients | Supports over 500 clients |
| **Software Features** | Cisco DNA support, CleanAir Pro, AI-enhanced analytics | ArubaOS with AI-powered RF optimization |
| **Security** | WPA3, Cisco TrustSec and identity-based access | WPA3, role-based access policies |
| **Management Options** | Cisco DNA Center, web GUI, CLI and third-party APIs | Aruba Central, CLI and local management |
| **Environmental Suitability** | Indoor and outdoor models available | Indoor model only |
| **Price** | RM3300 | RM2899.99 |

The major differences between the same devices from different brands often lie in performance, features, pricing and ecosystem integration. For example, Cisco provide advanced analytics, AI-driven optimization and higher client capacity while Aruba offers cost-effective solutions with simplified management and robust performance. In these two choices, Cisco is more meets our requirement for using in our faculty building although it is more expensive than Aruba. Therefore, Cisco is the better choice for us in this project.

**Projector**

For the projector used in this project, we have chosen **Viewsonic PA503X XGA** . The major difference between **Viewsonic PA503X XGA** with other different brands we can compare is **Epson PowerLite X49**.

| **Feature** | **Viewsonic PA503X XGA** | **Epson PowerLite X49** |
|---|---|---|
| **Brightness** | 3600 lumens | 3600 lumens |
| **Connectivity** | HDMI, USB, VGA, Wireless | HDMI, USB, VGA, Wireless |
| **Resolution** | XGA (1024*768) | XGA (1024*768) |
| **Contrast** | 22000:1 | 16000:1 |
| **Lamp Life** | 15000 hours | 12000 hours |
| **Audio** | 2W | 5W |
| **Projected Distance** | 1.189 - 13.106 m | 0.89-9.12 m |
| **Features** | Technology Digital Light Processing, sharper images and compact size | Moderator function, 3LCD technology, better for large images and energy efficient |
| **Price** | RM1799 | RM2296 |

Based on the comparison above, the ViewSonic PA503X XGA s a great choice for the classroom since it has a higher lumens brightness and contrast ratio compare to Epson PowerLite X49, it ensures clear and crisp images even in bright environments. The longer projection distance allows flexible placement within the room. Priced at RM1799, it offers excellent value for money. While the Epson PowerLite offers advanced features like enhanced audio and 3LCD technology, the ViewSonic PA503X XGA is more cost-effective and ideal for classrooms where flexibility and budget are important.

**Desktop**

For the desktop used in this project, we have chosen **HP Elite Tower 800 G9 Desktop**. The major difference between **HP Elite Tower 800 G9 Desktop** with other different brands we can compare is **Lenovo ThinkStation P3 Tower**.

| | **HP Elite Tower 800 G9 Desktop** | **Lenovo ThinkStation P3 Tower** |
|---|---|---|
| **Operating System** | Windows 11 Pro | Windows 11 Pro |
| **Processor** | Intel® Core™ i5-13500 (up to 4.8 GHz with Intel® Turbo Boost Technology, 24 MB L3 cache, 14 cores, 20 threads) | 13th Generation Intel® Core™ i5-13400 Processor (E-cores up to 3.30 GHz P-cores up to 4.60 GHz) |
| **Ports** | 1 headphone/microphone combo; 4 SuperSpeed USB Type-A 10Gbps signaling rate; 1 SuperSpeed USB Type-C® 20Gbps signaling rate | USB-C (10Gbps), USB-A (5Gbps and 10Gbps), HDMI 2.1 TMDS, dual DisplayPort™ 1.4, Ethernet (RJ45), headphone/mic combo, microphone input, line-out, optional SD card reader, optional expansion cards, and a power input. |
| **Graphics** | Intel® UHD Graphics 770 | Integrated Graphics |
| **Memory and Storage** | 8 GB memory; 512 GB SSD storage | 8 GB DDR5-4400MHz (UDIMM) memory; 256 GB SSD M.2 2280 PCIe Gen4 TLC Opal |
| **Price** | RM 4652 | RM 4079 |

Based on the comparison above, the **HP Elite Tower 800 G9** is more suitable for the faculty due to its superior performance and features. It is powered by an Intel Core i5-13500 processor with a higher clock speed and larger cache which will make it better equipped for multitasking and demanding tasks. The system includes Intel UHD Graphics 770, offering slightly better integrated graphics performance. It provides double the capacity of Lenovo ThinkStation P3 with 512GB of SSD storage for handling applications and data. Moreover, its SuperSpeed USB Type-C port with a 20Gbps signaling rate ensures faster data transfer for modern peripherals. Although its slightly more expensive, the HP Elite Tower 800 G9 offers better overall value and making it the optimal choice for academic and administrative use.

**Smartboard**

For the smartboard used in this project, we have chosen **Huawei Ideahub 2**. The major difference between **Huawei Ideahub 2** with other different brands we can compare is **Canon smart interactive board**.

| **Feature** | **Mediabridge Ethernet Cable** | **Ultra Clarity Cables Cat6** |
|---|---|---|
| **Display Size** | Available in 65", 75", 86" | Available in 70" |
| **Resolution** | 4K Ultra HD (2840*2160) | 4K |
| **Operation System** | HarmonyOS | Android and Windows |
| **Connectivity** | Wi-Fi 6, Bluetooth, USB and HDMI | Wi-Fi, Bluetooth and USB |
| **RAM** | Up to 8GB | - |
| **Storage** | 128GB SSD | - |
| **Camera** | 4K Video Camera | 4K Video Camera |
| **Latency** | 16ms Ultra Low Latency | - |
| **Audio Features** | Voice tracking, auto framing and acoustic baffle for superior audio quality | Speaking-tracking and auto-framing |
| **Unique Features** | BYOM (Bring Your Own Meeting), Intelligent Whiteboard | OCR (Smart Optical Character Recognition) |
| **Price** | RM15000-20000 | RM20000-25000 |

Based on the table above, the Huawei IdeaHub S2 may be the better option due to its advanced features, versatile connectivity options, and budget-friendly price. Its ultra-low writing latency and intelligent whiteboard functions make it perfect for interactive teaching and presentations. Additionally, the high-definition video and superior audio quality enhance virtual lectures and meetings, ensuring seamless and engaging learning experience. Although Canon Smart Interactive Board offers these capabilities, but it has a higher price compared to Huawei IdeaHub S2. Huawei IdeaHub S2's flexibility display sizes also cater to different room sizes, further adding to its suitability for educational environments.

**Cat6 Ethernet Cable**

For the Cat6 Ethernet Cable used in this project, we have chosen the **Mediabridge Ethernet Patch Cable Cat6**. The major difference between **Mediabridge Ethernet Patch Cable** with other different brands we can compare is the **Ultra Clarity Cables Cat6.**

| **Feature** | **Mediabridge Ethernet Cable** | **Ultra Clarity Cables Cat6** |
|---|---|---|
| **Build Quality** | Premium PVC jacket with gold-plated connectors for durability and corrosion resistance. | Durable jacket material with snagless connectors for long-lasting performance. |

| | | |
|---|---|---|
| **Cable Length Options** | Available in 6 ft, 10 ft, 25 ft, 50 ft, 100 ft, and bulk 1000 ft rolls. | Available in 6 ft, 10 ft, 15 ft, 25 ft, 50 ft, and longer lengths up to 100 ft. |
| **Performance Standard** | Meets or exceeds Cat6 performance standards, supporting speeds up to 1 Gbps with a bandwidth of 250 MHz. | Meets Cat6 standards, supports speeds up to 1 Gbps, and offers up to 250 MHz bandwidth. |
| **Connector Type** | RJ45 connectors with snagless design for easy plugging/unplugging and secure connections. | RJ45 connectors with gold-plating for reduced signal loss and enhanced connectivity. |
| **Cable Type** | UTP (Unshielded Twisted Pair), suitable for most general-purpose networking needs. | UTP (Unshielded Twisted Pair), ideal for home, office, and lab environments. |
| **Color Options** | Commonly available in white, black, and blue. | Available in multiple colors like black, blue, gray, and white. |
| **Price** | RM 45 - RM 65 for 6 ft, RM 135 - RM 225 for 100 ft. Bulk 1000 ft: RM 365 - RM 550. | RM 45 - RM 90 for 6-15 ft, RM 135 - RM 225 for longer lengths (up to 100 ft). |
| **Ease of Use** | Flexible and easy to manage; compatible with Ethernet-enabled devices, including routers, switches, and PCs. | Flexible cable with snagless boots, suitable for all Cat6-compatible devices. |
| **Ideal Use Case** | Suitable for general networking in labs, offices, or homes where high-speed, reliable connections are needed. | Good for university labs, home offices, and general networking applications. |
| **Unique Features** | Bulk 1000 ft roll is an affordable option for large-scale wiring projects. | Slightly more color options for cable organization and aesthetics. |

Both cables perform to Cat6 standards, making them comparable in speed and bandwidth. However, the Mediabridge offers a slight advantage in pricing for bulk purchases, as 1000 ft rolls are more economical than buying multiple shorter lengths for faculty labs use. Mediabridge is well-suited for large-scale networking projects due to its bulk option, while Ultra Clarity is preferable for smaller setups that require a small range of lengths with different colours. For our project, the Mediabridge Ethernet Cable is the recommended better choice to connect numerous devices or plan extensive wiring across labs and areas, with the bulk roll option being an added benefit.

### Fiber Optic Cables

For the Fiber Optic Cable used in this project, we have chosen the **SC APC Patch Cable, OS2**. The major difference between **SC APC Patch Cable, OS2** with other different brands we can compare is the **LC UPC Patch Cable, OM4.**

| | **LC UPC Patch Cable, OM4** | **SC APC Patch Cable, OS2** |
|---|---|---|
| **Fiber count** | 2 fibers | 1 fibers |

| | | |
|---|---|---|
| **Wavelength** | 850/1300nm | 1310/1550nm |
| **Insertion Loss** | <=0.3dB | <=0.3dB |
| **Return Loss** | >=30dB | >=60dB |

The **SC APC Patch Cable, OS2** is more suitable choice for the faculty due to its compatibility with long-distance single-mode fiber applications, making it ideal for connecting buildings or linking to an ISP. It supports high-speed data transmission over extended distances with its wavelength range to ensuring reliable connectivity for the faculty's infrastructure. Moreover, its high return loss indicates excellent signal quality. Overall, the SC APC Patch Cable aligns better with the faculty's need for scalable and robust networking.

### 4.3.3 Question and Answer in Choosing Networking Devices

1. Are you surprised by the price? How were you surprised?

The total cost of RM1,024,812 for the network setup is not surprising because we have chosen the best devices to ensure performance, reliability and scalability. For example, premium equipment like Cisco Catalyst 9130AX access points and Dell PowerEdge R750 servers were selected to meet the demands of high-speed connectivity, large user capacity and advanced functionality. High-quality devices are essential for supporting a stable and efficient network infrastructure, especially in an educational environment where labs, classrooms and offices rely on uninterrupted access.

Additionally, the budget was calculated to balance quality and cost-effectiveness. Every component, from routers and switches to patch panels and cables, was evaluated for its ability to meet the requirements without unnecessary overspending. By investing in reliable and durable devices while maintaining financial discipline, the setup ensures both immediate performance and long-term sustainability, making the overall cost reasonable and well-justified.

2. Have you ever considered cost as a factor for choosing networking devices?

When choosing networking devices, cost was not our main focus but we will always survey for the most affordable and quality devices. Our priority was to identify devices that best met our project needs and enhanced network efficiency within our budget. We thoroughly assess different options and select those that provide the ideal balance of performance, affordability, and alignment with our project objectives.

However, the cost is still an important consideration in selecting networking devices, even though it wasn't the primary focus. Here's a detailed and explanation why we conclude that the cost is partially one of the factors for choosing networking devices:

1. Balancing Cost with Functionality

We have conducted surveys and comparisons across brands to identify devices that provide us the best balance of affordability and performance. For example, the router, Ubiquiti EdgeRouter 12 was selected for its advanced features like 12 Ethernet ports and higher throughput while being cost-effective for large-scale setups compared to alternatives like the Cisco Catalyst, which is significantly more expensive. For the patch panels, the BELNET CAT6 24-Port Patch Panel was chosen over the Panduit Mini-Com® due to its lower price (RM 378 vs. RM 627.13) while still meeting project requirements for cable management and functionality.We ensure that devices fulfilled project objectives like supporting high-speed networks or handling high user capacity without exceeding the budget.

2. Strategic Cost Management

We make sure the devices we choose stay within the budget. For instance, the Cat6 Ethernet Cable, Bulk Mediabridge 1000-foot rolls were highlighted as more economical for extensive wiring across labs. For the Smartboards, The Huawei IdeaHub S2 was selected over Canon's option due to its lower price (RM 15,000 vs. RM 20,000-25,000), even though both provided interactive teaching capabilities. While cost wasn't the only consideration, it helped narrow choices to devices that offered value while avoiding unnecessary premium features.

3. Understanding the Project's Scope

With the requirement to equip multiple labs and offices, the cost became crucial in scaling the network effectively. Devices with scalability and high utility per cost unit were favoured, such as the Cisco Catalyst 9130AX series Access Points. These were selected despite being slightly more expensive than Aruba options because they supported a larger number of clients (over 1,000) and offered additional features like AI-enhanced analytics.

4. Long-Term Cost Implications

Beyond the initial purchase price, your team considered devices that minimized long-term costs. The Dell PowerEdge R750 server was chosen despite its higher upfront cost because of its superior performance, scalability, and management features, which reduce maintenance costs over time. While for the Fiber Optic Cables, the SC APC Patch Cable was selected for its high return loss (>=60dB), ensuring minimal signal loss and reliable long-distance data transmission, reducing the need for frequent replacements or troubleshooting.

While cost wasn't your primary focus, we acknowledge that it played a key role in ensuring the project stayed within budget. We compared multiple options and strategically opted for devices that achieved the best balance of quality, functionality, and affordability.

### 4.3.4 Reflection on Task 3

Task 3 provided valuable insights into the importance of selecting the right networking devices while balancing performance, functionality and cost-effectiveness. Initially, we was surprised by the high cost of networking equipment but through extensive research and comparisons, we realised that investing in high-quality and sccalable devices is crucial for long-term efficiency. We learned to evaluate different brands and models, considering factors such as network capacity and future scalability, ensuring that our selections aligned with the faculty's needs.

Another key takeaway from this task was the process of cost management. While our priority was network performance and reliability, we had to strategically allocate the budget to avoid unnecessary overspending. By analysing various devices such as routers, switches, patch panels and cabling, we identified cost-effective options that met both technical and financial constraints. This experience helped us understand the trade-offs between affordability and quality in network planning.

Overall, Task 3 has enhanced our ability to do research, compare and justify network hardware choices based on both technical specifications and budget considerations. This hands-on experience has strengthened our practical knowledge of network setup.

## 4.4 Task 4 LAN and WAN Connection

### 4.4.1 Identify Work Areas on Floor Plan

<u>Work Area at Ground Floor</u>


Figure 4.4.1: Work Area at Ground Floor

The Work Area 1 and 3 is General Lab which is 14m in length and 10m in width. This lab is to provide a space for students to work on general computing tasks, coursework and software-based projects. Each lab is equipped with a total of 31 PC workstations including the one at the lecturer's table. Consequently, the total of 24-port switches needed by the general labs is three switches to connect to all the workstations.

Next, for Work Area 2 that shown in Figure 4.4.1, is a Cisco Networking Lab which has a purpose to offer hands-on experience in network setup, configuration and troubleshooting, preparing students for networking careers. This lab also consists of 30 workstations and a total of 5 server racks for students to learn about networking setup. In this lab, there is also a projector that can show teaching material easily.

At the bottom right of Figure 4.4.1, there is work area 4 which is Embedded Lab. This lab is built for enabling students to learn hardware-software integration and embedded system development, focusing on IoT and real-time applications. It contains 32 workstations including lecturer's PC and projector for easy teaching in the class.

Work Area of First Floor

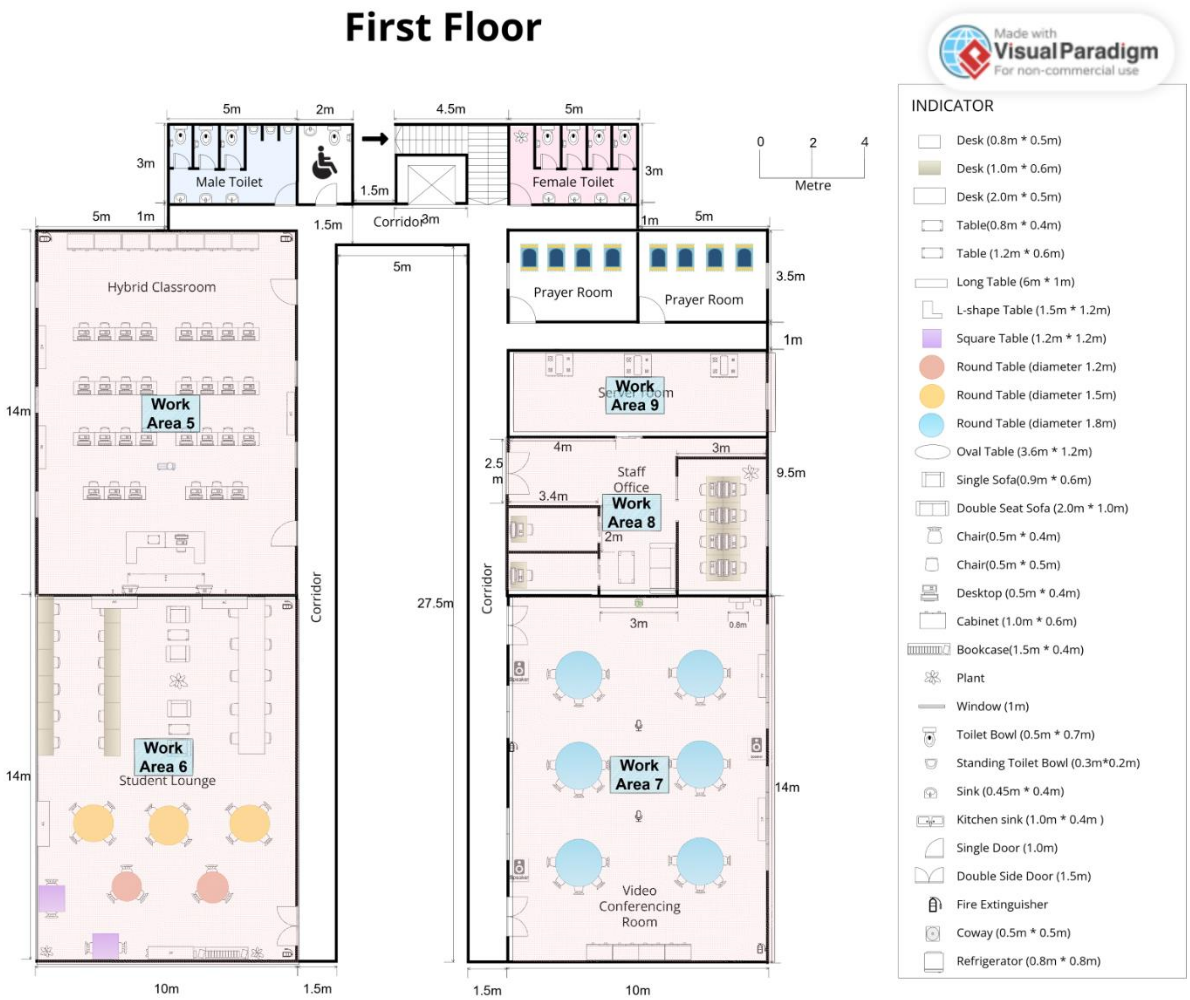



Figure 4.4.2: Work Area at First Floor

Work area 5 shown in Figure 4.4.2 is a hybrid classroom which is designed for interactive learning that combines physical and virtual sessions. It is equipped with 31 PC workstations and advanced technology such as projector, whiteboard and sound system, ensuring an engaging learning environment for both in-person and remote participants.

In Figure 4.4.2, the bottom left shows a student lounge space for students to relax and collaborate. It features comfortable seating options such as sofas, round tables for group discussions and a cozy environment to encourage informal interactions and socialization.

Meanwhile, work area 7 is a video conferencing room which serves as a dedicated space for virtual meetings, presentations and collaborative sessions, supporting remote connectivity with high-speed internet and audio-visual tools. It is equipped with round tables, ergonomic chairs and high-quality video conferencing systems.

Next, work area 8 shown in Figure 4.4.2 is a staff office which is designed for faculty and administrative tasks. It includes 8 PC workstations in common space while each small workspace has 1 PC workstation to ensure a professional workspace for staff, enhancing productivity and organization.

Lastly, work area 9 is a server room dedicated to housing and managing critical IT infrastructure. It includes 3 server racks and cooling systems to maintain optimal temperature and performance. This secure area is designed to ensure reliable connectivity and data management for the entire floor.

## 4.4.2 Distribution and Connection of Devices

<u>Network Distribution Diagram</u>

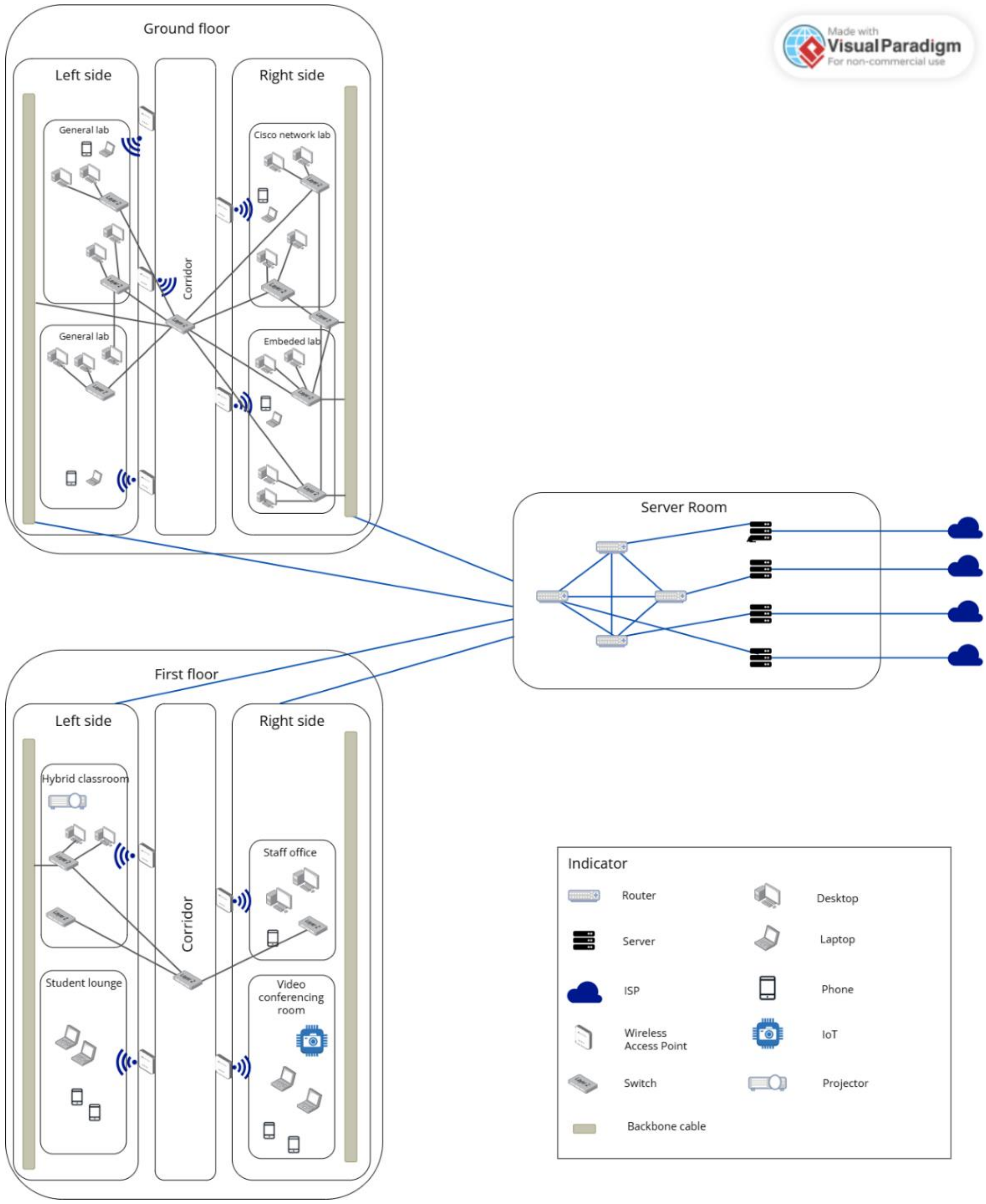



Figure 4.4.3 Network Distribution Diagram

In Figure 4.4.3, the diagram illustrates the network's layout and connectivity for both the ground and first floors of the building, ensuring seamless integration of wired and wireless devices. On each floor, switches are deployed to connect wired devices such as desktops and IoT devices in areas like general labs, Cisco Network Lab, Embedded Lab, hybrid classroom, staff office and video conferencing room. Wireless access points are strategically places to provide Wi-Fi coverage, enabling mobile and wireless device connectivity. Each switch serves as a local hub, consolidating connections and maintaining efficient data flow within its designated area.

All switches on the ground and first floors are linked to the server room via backbone cables. Which ensures high-speed and reliable data transfer between floors and core switch. In the server room, servers are connected to switches for managing computing tasks, data storage and application hosting. These switches are further linked to a router, which connects to the ISP for internet access. This hierarchical structure ensures a centralized yet flexible network arrangement, supporting both high-performance wired connections and robust wireless communication throughout the building.

For a clearer depiction of the network distribution on the ground and first floors, refer to figures 4.4.4 and 4.4.5. The diagram outlines how wireless access points, switches and devices in various sections of the building connect through backbone cables to the switches and router housed in the server room. The connections are simplified to highlight the overall structure, while Figure 4.4.6 provides a detailed illustration of the server room.

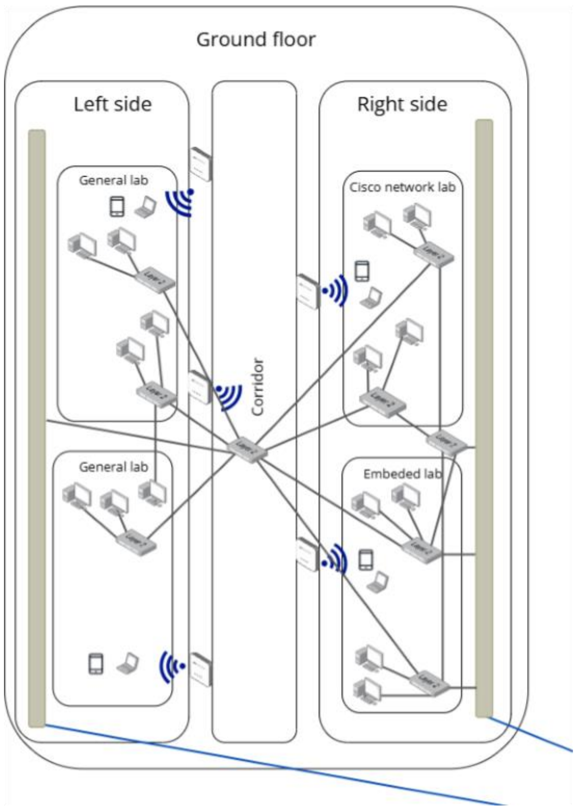



Figure 4.4.4 Network Distribution Diagram – Ground Floor

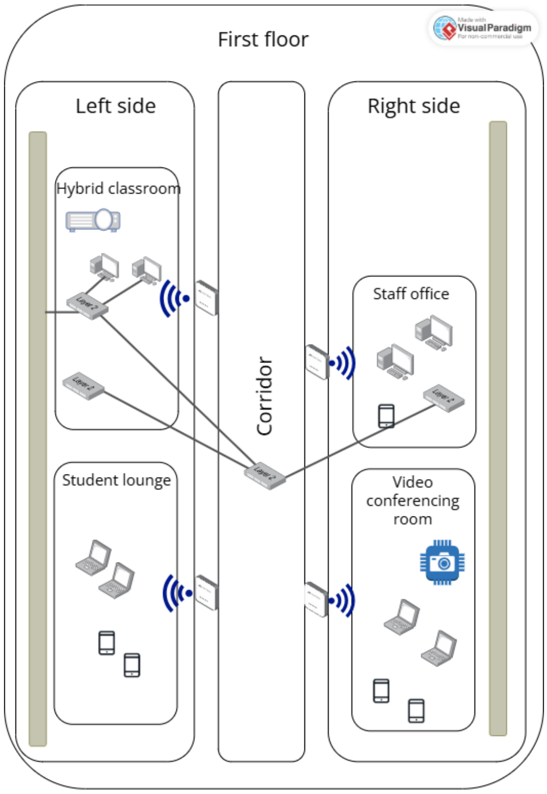



Figure 4.4.5 Network Distribution Diagram – First Floor

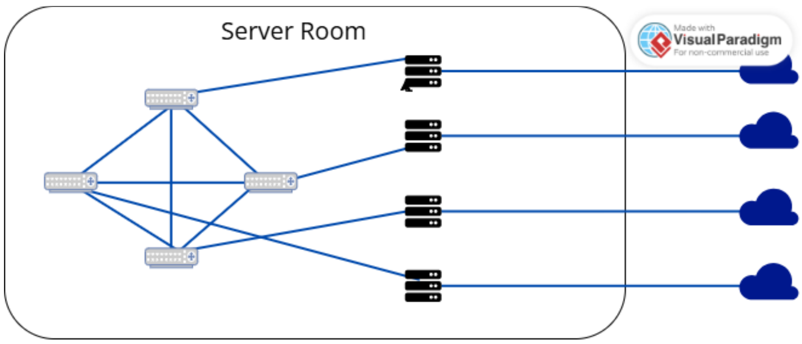



Figure 4.4.6 Network Distribution Diagram – Server Room

The depicted diagram illustrates the network distribution across the ground floor, first floor and server room with an emphasis on the router and server configuration. In this design, four routers are integrated into the network within the server room, each independently connected using fiber optic cables. This configuration is intended to prevent a complete network outage in the event of a single router failure.

To enhance network resilience, the routers are arranged in a mesh topology which allows seamless functionality even if one router encounters issues. Across the building, a single switch is deployed in most rooms to establish localized subnets, ensuring stability within each subnet. Moreover, one switch is redundantly connected to two routers which provide a failover mechanism to maintain network operations during potential disruptions or emergencies. This strategic setup ensures robust connectivity and fault tolerance across all network layers.

<u>Floor Diagram</u>

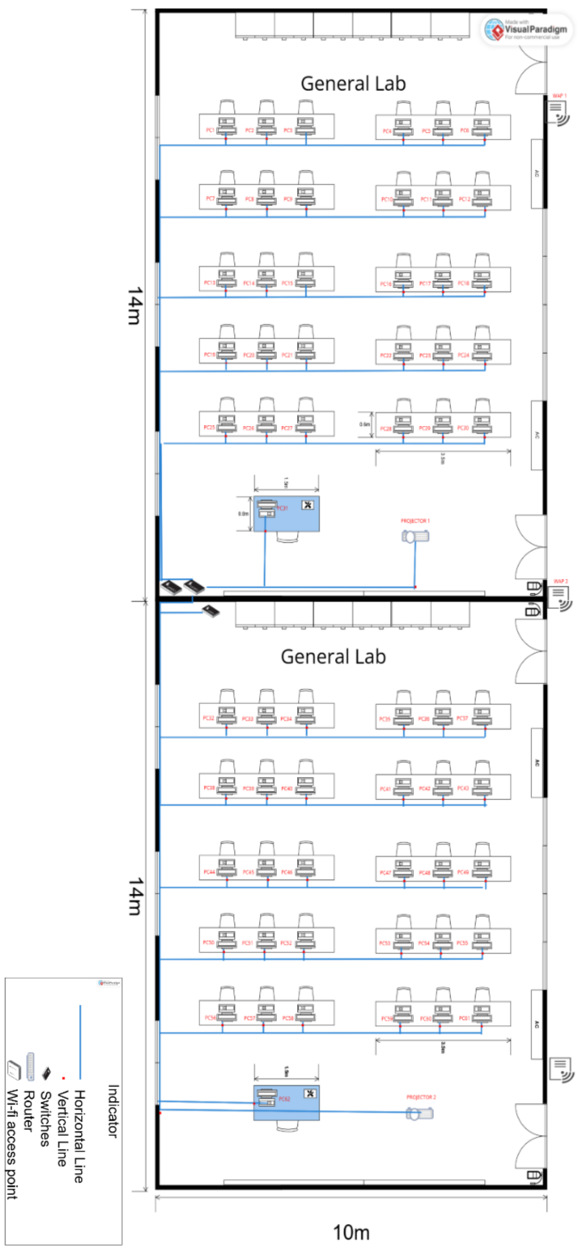



Figure 4.4.7 Cable Distribution Diagram – General Labs

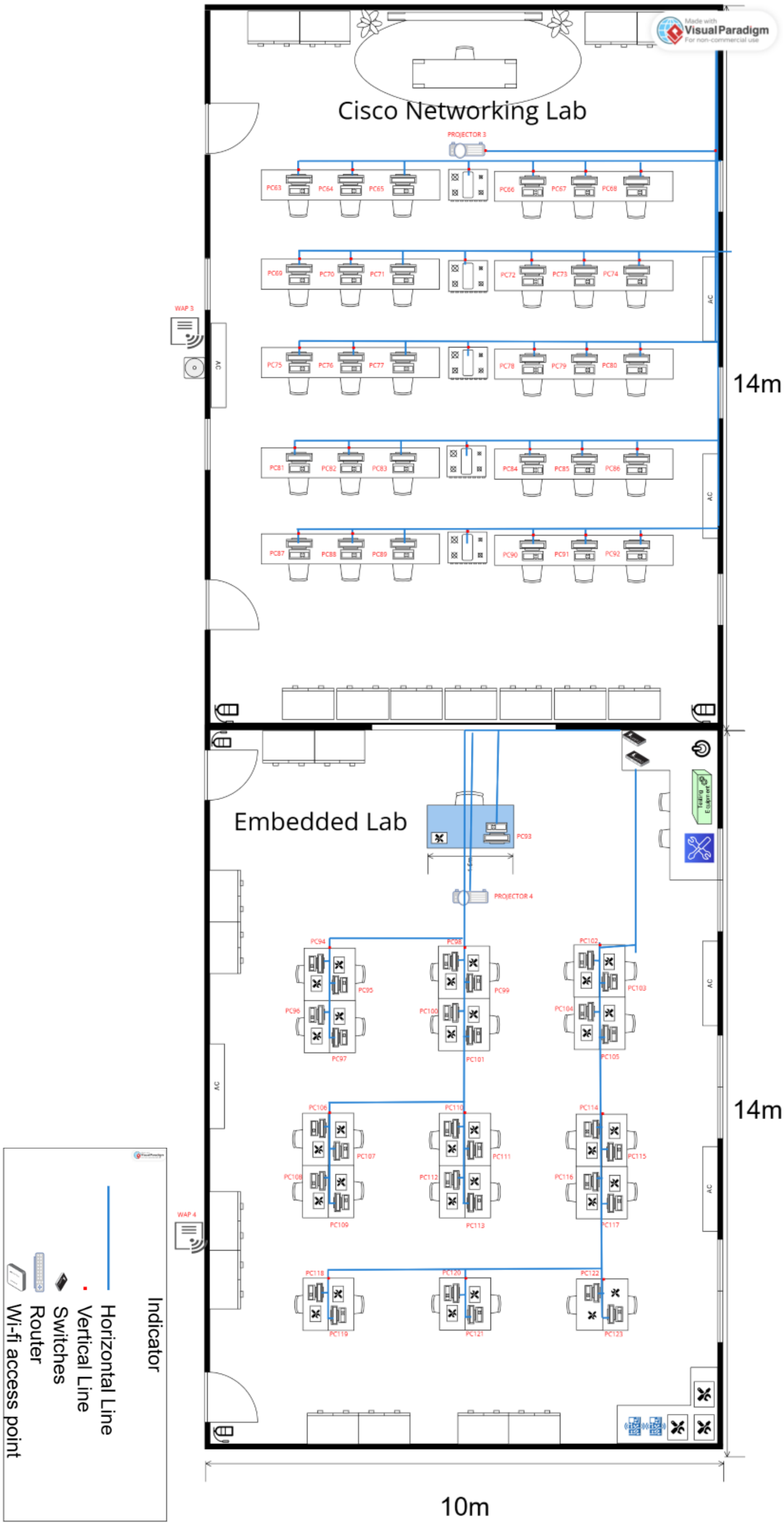



Figure 4.4.8 Cable Distribution Diagram – Cisco Networking Lab and Embedded Lab

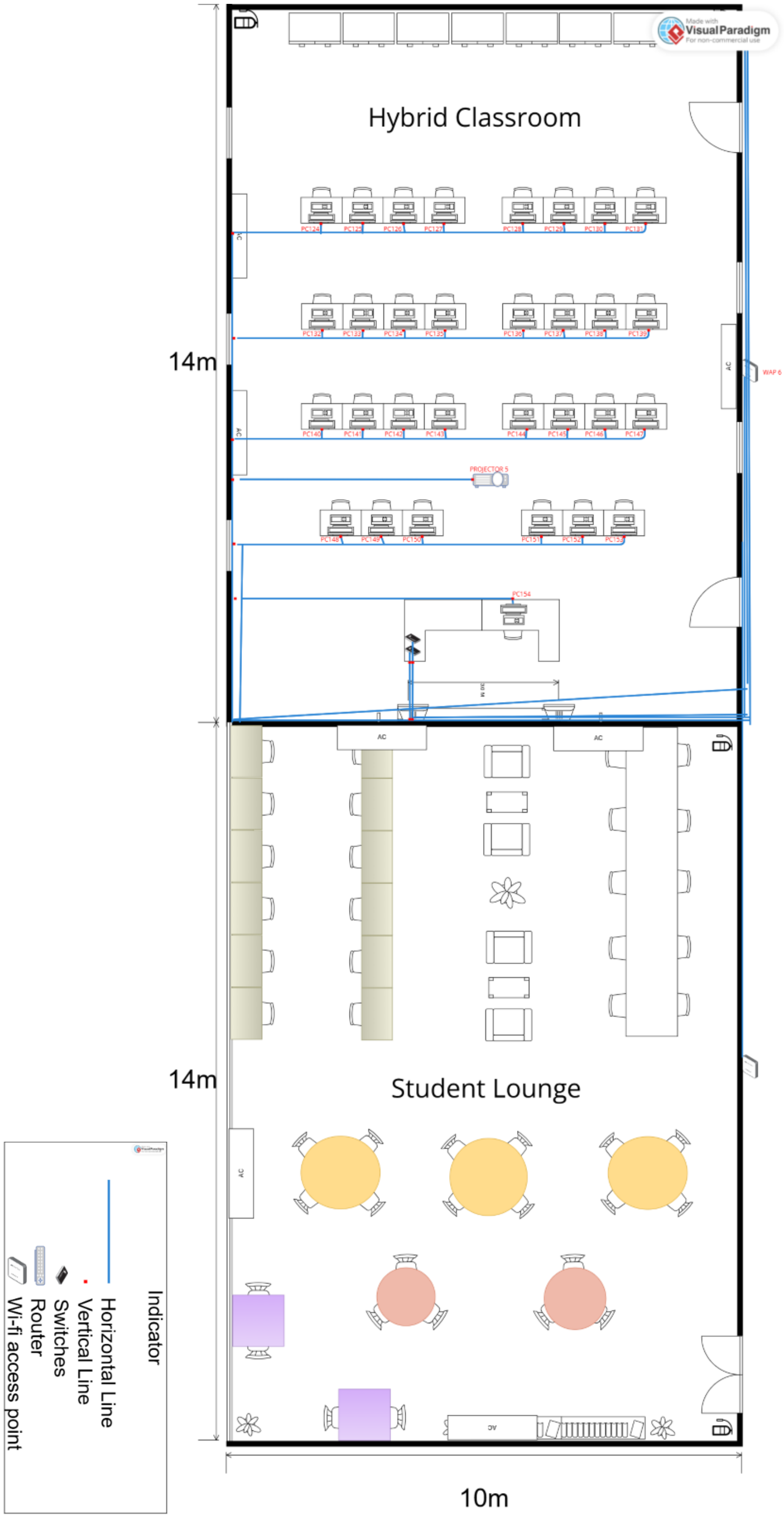



Figure 4.4.9 Cable Distribution Diagram – Hybrid Classroom and Student Lounge

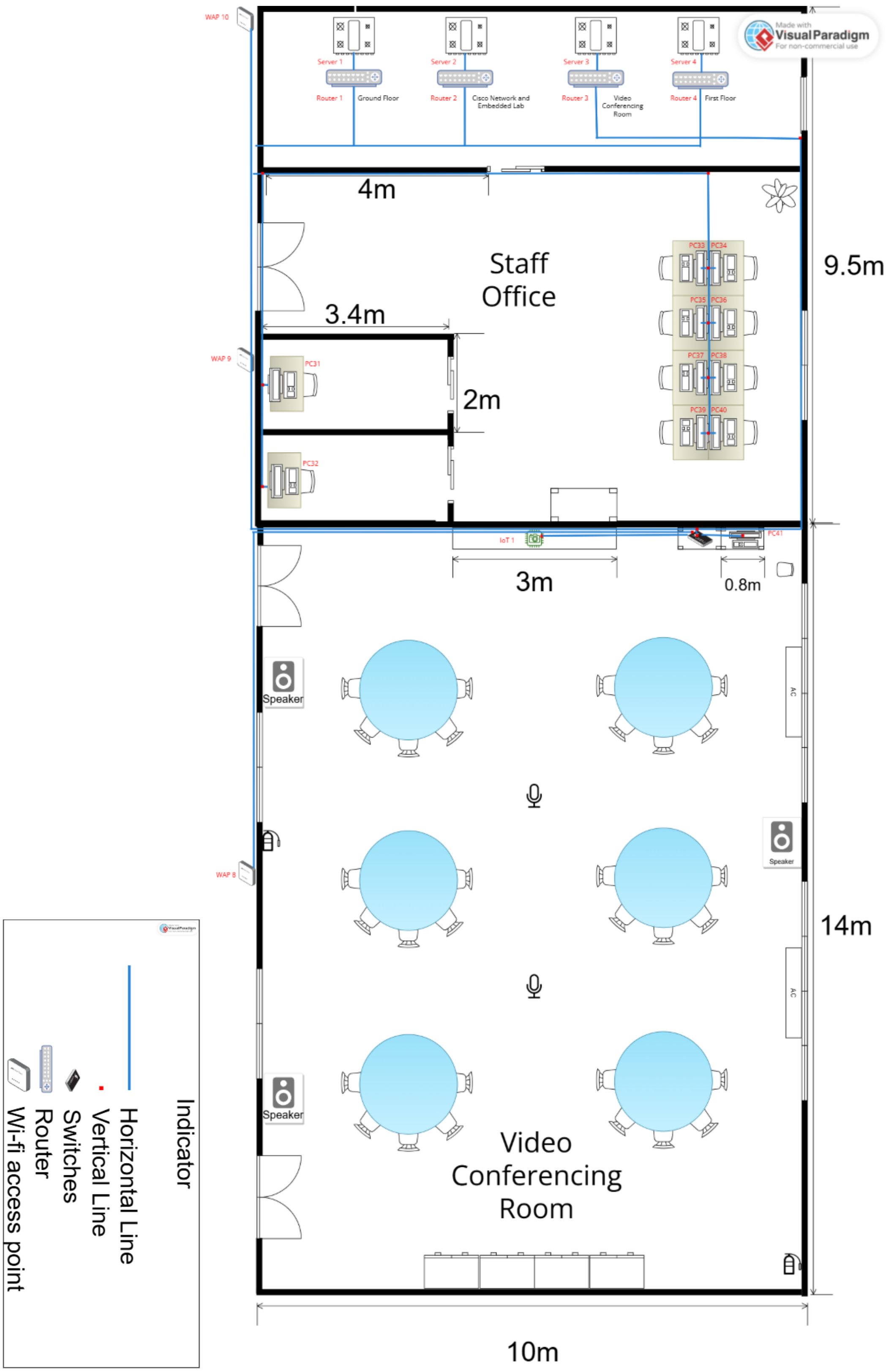



Figure 4.4.10 Cable Distribution Diagram –Staff Office, Server Room and Video Conferencing Room

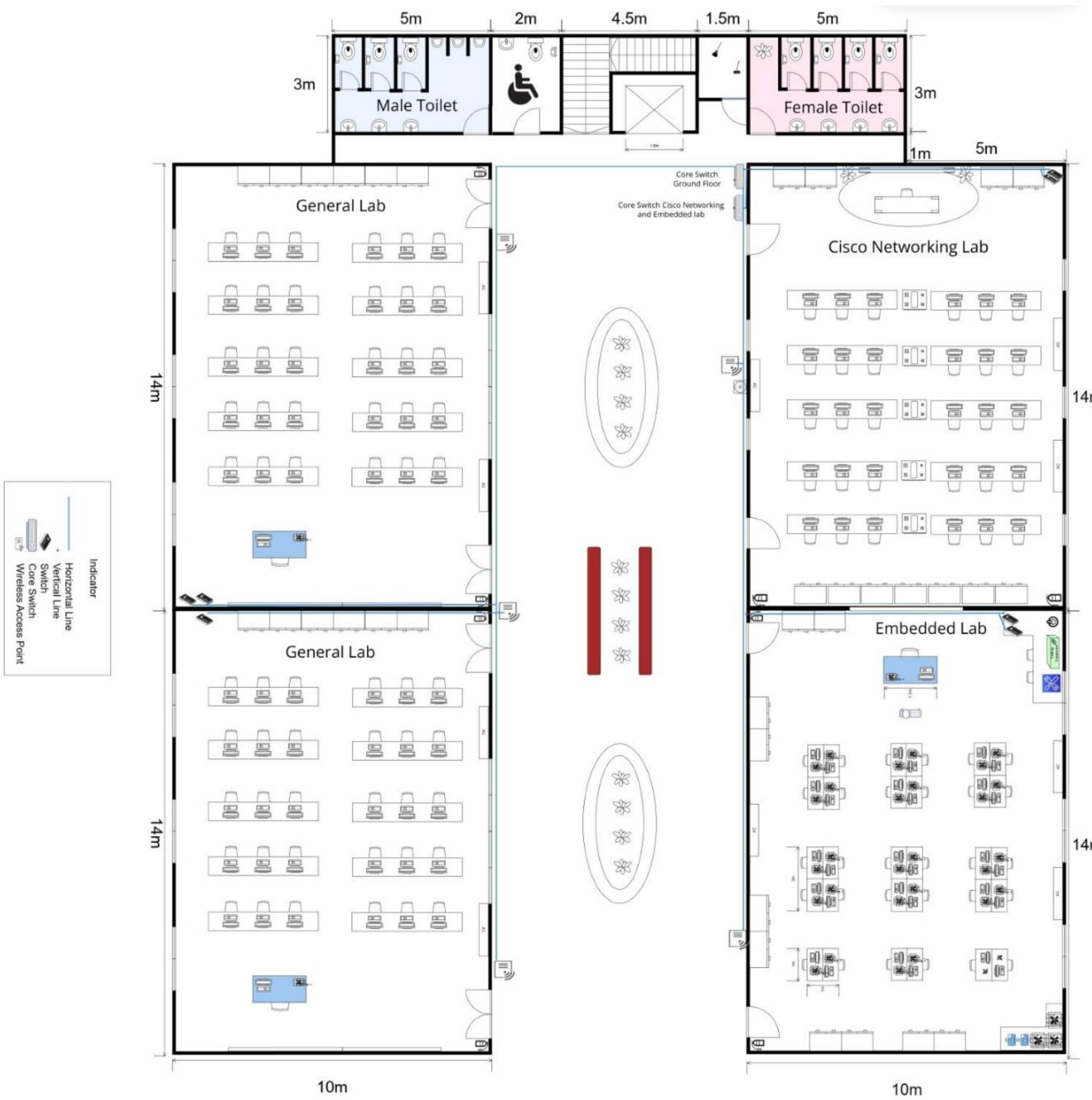



Figure 4.4.11 Cable Distribution Diagram – Ground Floor

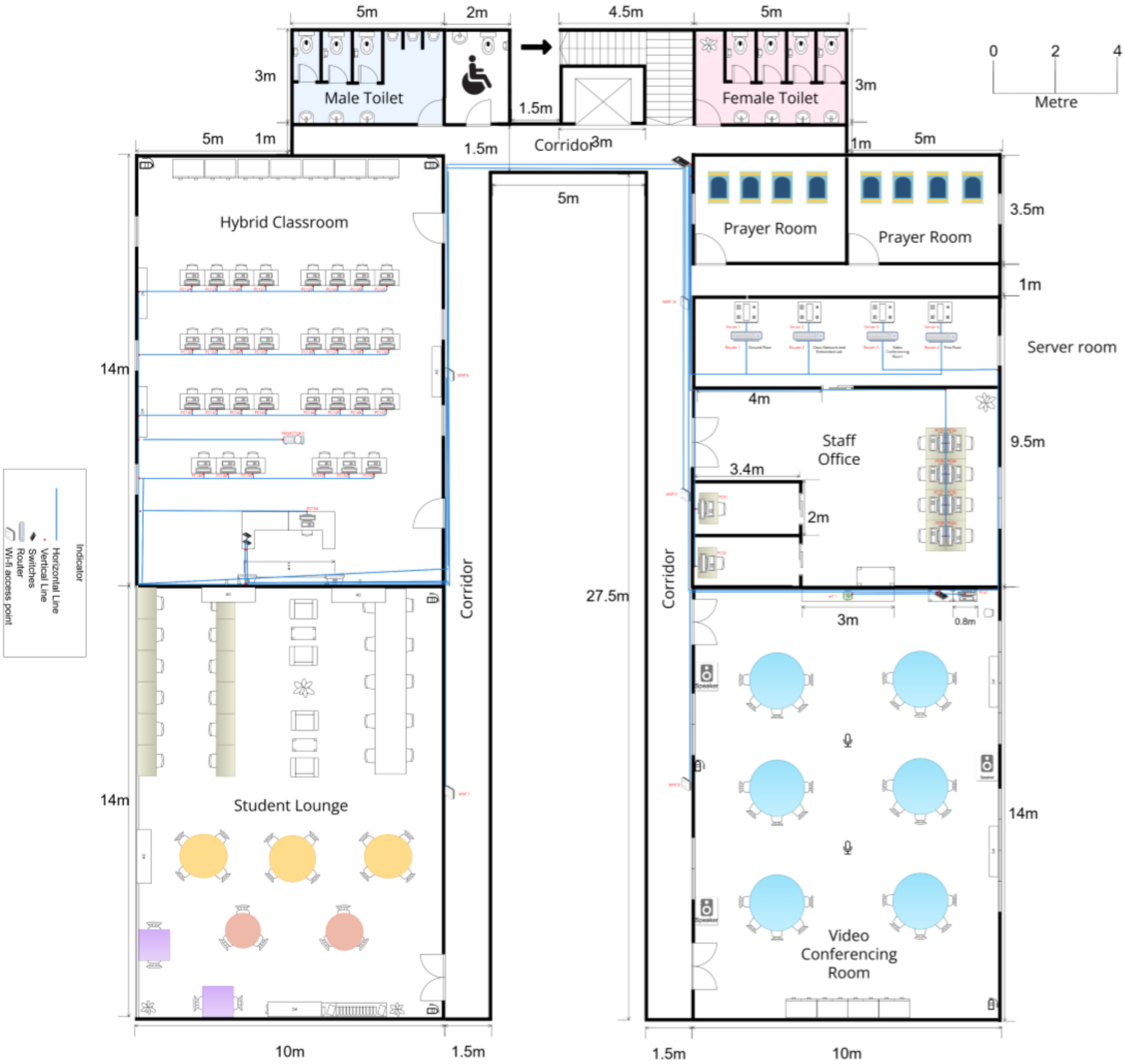



Figure 4.4.12 Cable Distribution Diagram – First Floor

The floor diagrams illustrate the network connections within the faculty building, showing how switches on each floor are interconnected and linked via backbone connections. Each floor has a core switch that consolidates connections from all individual switches on that floor. The core switches are then linked to dedicated routers, establishing a robust and efficient network. Additionally, key areas such as the Embedded Lab, Cisco Networking Lab, and the Video Conferencing Room are highlighted in the diagrams, showing their direct or indirect connections to the routers.

### 4.4.3 Number of Connections, Patch Cords and Switch Ports Needed

| Area | Number of Switch Ports Needed |
|---|---|
| General Lab 1 | 36 |
| General Lab 2 | 34 |
| Cisco Network Lab | 39 |
| Embedded Lab | 35 |
| Hybrid Classroom | 36 |
| Video Conference Room | 5 |
| Staff Office | 10 |
| Server Room | 5 |
| Total | 200 |

Table 4.4.1: Number of Switch Ports Needed

| Area | Number of Switch Ports Needed |
|---|---|
| General Lab 1 | 36 |
| General Lab 2 | 34 |
| Cisco Network Lab | 39 |
| Embedded Lab | 35 |
| Hybrid Classroom | 36 |
| Video Conference Room | 5 |
| Staff Office | 10 |
| Server Room | 5 |
| Total | 200 |

Table 4.4.2: Number of Patch Cords Needed

| **Area** | **Number of Switch Ports Needed** |
|---|---|
| General Lab 1 | 72 |
| General Lab 2 | 68 |
| Cisco Network Lab | 78 |
| Embedded Lab | 70 |
| Hybrid Classroom | 72 |
| Video Conference Room | 10 |
| Staff Office | 20 |
| Server Room | 10 |
| Total | 400 |

Table 4.4.3: Number of RJ45 Connector Needed

### 4.4.4 Identify Cable Types and Length

| Area | Length (m) | | Total Length (m) | Total Price of Cable Used (RM) |
|---|---|---|---|---|
| | Horizontal | Vertical | | |
| Ground Floor | | | | |
| General Lab 1 | 184.5 | 27.3 | 211.8 | 1129.6 |
| General Lab 2 | 191.5 | 26.5 | 218 | 1162.67 |
| Cisco Network Lab | 191 | 29.5 | 220.5 | 1176 |
| Embedded Lab | 74.1 | 16.5 | 90.6 | 483.2 |
| Ground Floor Room Total | | | 740.9 | 3951.47 |
| First Floor | | | | |
| Hybrid Classroom | 82.8 | 47.05 | 129.85 | 692.53 |
| Video Conference Room | 21.80 | 5.00 | 26.80 | 142.94 |
| Server Room | 17.40 | 5.60 | 23.00 | 122.67 |
| Staff Office | 27.2 | 13.1 | 40.3 | 214.93 |
| First Floor Room Total | | | 213.75 | 1173.07 |

| Path | Length(m) | Price of cables used (RM) |
|---|---|---|
| Ground Floor | | |
| Server room to Core Switch Ground Floor | 15 | 80 |
| Server room to Core Switch Cisco Networking Lab and Embedded Lab | 15 | 80 |
| Wifi Access Point to switches | 59 | 314.7 |

| | | |
|---|---|---|
| Switches to Core Switch | 117 | 624 |
| Ground Floor Room Total | 206 | 1098.7 |
| Total - Ground Floor | | |
| Total length of wire to connect the rooms (m) | 206 | 1098.7 |
| Total length of wire to connect end devices to switches in rooms (m) | 740.9 | 3951.47 |
| Total length of wire needed (m) | 946.9 | 5050.17 |
| First Floor | | |
| Wi-Fi Access Point to Switches | 37.4 | 59.84 |
| Hybrid Classroom to core switch | 46.4 | 74.24 |
| Video Conferencing Room to core switch | 14 | 22.4 |
| Staff Office to Core Switch | 18 | 28.8 |
| Server Room to Core Switch | 7.2 | 11.52 |
| First Floor Total | 123 | 196.80 |
| Total First Floor | | |
| Total length of wire to connect the room (m) | 123 | 196.80 |
| Total length of wire to connect end devices to switches in rooms (m) | 213.75 | 1173.07 |
| Total length of wire needed (m) | 336.75 | 1369.87 |

| | | |
|---|---|---|
| Total (Ground Floor + First Floor) (m) | 1283.65 | 6420.04 |

Choosing the right cables is important for good connectivity. After discussing, we'll use both CAT6 twisted cables and fiber optic cables in the building.

We'll use CAT6 twisted cables for end devices connections in labs and classroom such as desktops, projector and other network end user devices. These cables can handle high-speed data up to 10 gigabits per second since they have a high bandwidth. CAT6 cables have thicker copper wires that help with heat dissipation, making them good for Power over Ethernet (PoE), which reduces cabling time and costs. The specific type of CAT6 chosen is Unshielded Twisted Pair (UTP), known for reducing electromagnetic interference caused by natural or human-made sources.

While fiber optic cables are designated for interconnecting routers within the server room and extending the connection to the Internet Services Provider (ISP). By utilizing light for data transmission, fiber optic cables facilitate high-speed data transfer while minimizing signal degradation. With a bandwidth capacity up to 10Gbps, these cables are well-suited for handling significant data volumes. Moreover, fiber optic cables are highly effective at transmitting signals over longer distances, which guarantees fast and reliable data transmission.

In summary, our building requires a specific cable length, including both CAT6 and fiberoptic cables. This will ensure seamless internal connectivity among devices within the building and a reliable connection to the Internet Service Provider (ISP).

### 4.4.5 Reflection Task 4

This task helped us gain a deeper understanding of network design and implementation. We learned how to structure LAN and WAN networks, ensuring efficient data flow and reliable connectivity. Understanding how to connect all the network devices allowed us to design a more stable and scalable network system. We explored cable management which included selecting the right cable types, estimating lengths, and planning routing paths for better performance and cost efficiency. We also gained insight into how switch ports, patch cords, and backbone connections play a crucial role in maintaining a strong and efficient network.

In addition, this task helped us develop skills in network planning and resource estimation. We learned how to calculate the required number of connections, cable lengths, and networking equipment based on real-world needs. Understanding the difference between CAT6 cables for local connections and fibre optic cables for high-speed data transfer allowed us to see how different technologies are applied depending on network requirements.

Overall, this task improved our knowledge of network architecture, hardware selection, and efficient resource management. It provided us with valuable practical experience in designing a well-structured, cost-effective, and reliable network infrastructure while considering both technical requirements and real-world constraints.

## 4.5 Task 5

### 4.5.1 Network Address Given

| Group | Network Address |
|---|---|
| 12.3 | 192.164.0.0/8 |

The given network address of our group is 192.164.0.0/8. It provides over 16 million IPs (2^24) which making it suitable for large-scale networks with thousands of devices. This ensures future scalability for massive expansion.

Network Address details:

| | |
|---|---|
| IP address (decimal) | 192.164.0.0 |
| IP address (binary) | 11000000 10100100 00000000 00000000 |
| Subnet mask (decimal) | 255.0.0.0 |
| Subnet Mask (binary) | 11111111 00000000 00000000 00000000 |

We divided the subnet mask into the network portion (8 bits) and host portion (24 bits). By using the AND operation of the IP address with the subnet mask, we identified the following:

| | |
|---|---|
| Network Address | 192.0.0.0 |
| Broadcast Address | 192.255.255.255 |
| Range of Usable Addresses | 192.0.0.1 – 192.255.255.254 |

### 4.5.2 Divide it in the best possible way for network

By using the Classless Inter-Domain Routing (CIDR) method, we decided to divide the network into separate subnets for each work area outlined in task 4, except for the server room and staff office which share the same subnet. A total of 8 subnets are needed to accommodate the design of our network. Thus, we need to borrow 3 bits from the host portion of the IP address, as

$2_3$ *23*= 8 subnets are necessary to meet the network's segmentation requirements.
By borrowing these 3 bits, the subnet mask changed form /8 to /11.

Network Portion = Original 8 bits + 3 borrowed subnet bits = 11 bits

Host Portion = Total 32 bits – 11 Network Portion bits = 21 bits

This approach ensures that each subnet and devices and device have sufficient IP addresses while maintaining future scalability and efficient resource allocation across the faculty's network infrastructure.

Divide subnet to 8 subnets:

| Subnet Network | Network portion | Subnet bits | Host portion | | IP address |
|---|---|---|---|---|---|
| 0 | 1100 0000 | 000 | 0 0000<br>0000 0000<br>0000 0000 | Network Address | 192.0.0.0 |
| | | | 1 1111<br>1111 1111<br>1111 1111 | Broadcast Address | 192.31.255.255 |
| 1 | 1100 0000 | 001 | 0 0000<br>0000 0000<br>0000 0000 | Network Address | 192.32.0.0 |
| | | | 1 1111<br>1111 1111<br>1111 1111 | Broadcast Address | 192.63.255.255 |
| 2 | 1100 0000 | 010 | 0 0000<br>0000 0000<br>0000 0000 | Network Address | 192.64.0.0 |
| | | | 1 1111<br>1111 1111<br>1111 1111 | Broadcast Address | 192.95.255.255 |
| 3 | 1100 0000 | 011 | 0 0000<br>0000 0000<br>0000 0000 | Network Address | 192.96.0.0 |
| | | | 1 1111<br>1111 1111<br>1111 1111 | Broadcast Address | 192.127.255.255 |
| 4 | 1100 0000 | 100 | 0 0000<br>0000 0000<br>0000 0000 | Network Address | 192.128.0.0 |
| | | | 1 1111<br>1111 1111<br>1111 1111 | Broadcast Address | 192.159.255.255 |
| 5 | 1100 0000 | 101 | 0 0000<br>0000 0000<br>0000 0000 | Network Address | 192.160.0.0 |
| | | | 1 1111<br>1111 1111<br>1111 1111 | Broadcast Address | 192.191.255.255 |
| 6 | 1100 0000 | 110 | 0 0000<br>0000 0000<br>0000 0000 | Network Address | 192.192.0.0 |
| | | | 1 1111<br>1111 1111<br>1111 1111 | Broadcast Address | 192.223.255.255 |
| 7 | 1100 0000 | 111 | 0 0000<br>0000 0000<br>0000 0000 | Network Address | 192.224.0.0 |
| | | | 1 1111<br>1111 1111 | Broadcast Address | 192.255.255.255 |

| | | | | | |
|---|---|---|---|---|---|
| | | | 1111 1111 | | |

Distribution of subnet for different work area:

| Subnet | Work Area | Network Address | Broadcast Address | Range of usable address | Custom subnet mask |
|---|---|---|---|---|---|
| 0 | Server Room and Staff Office | 192.0.0.0 | 192.31.255.255 | 192.0.0.1 - 192.31.255.254 | /11 255.224.0.0 |
| Ground Floor | | | | | |
| 1 | General Lab 1 | 192.32.0.0 | 192.63.255.255 | 192.32.0.1 - 192.63.255.254 | /11 255.224.0.0 |
| 2 | General Lab 2 | 192.64.0.0 | 192.95.255.255 | 192.64.0.1 - 192.95.255.254 | /11 255.224.0.0 |
| 3 | Cisco Networking Lab | 192.96.0.0 | 192.127.255.255 | 192.96.0.1 - 192.127.255.254 | /11 255.224.0.0 |
| 4 | Embedded Lab | 192.128.0.0 | 192.159.255.255 | 192.128.0.1 - 192.159.255.254 | /11 255.224.0.0 |
| First Floor | | | | | |
| 5 | Hybrid Classroom | 192.160.0.0 | 192.191.255.255 | 192.160.0.1 - 192.191.255.254 | /11 255.224.0.0 |
| 6 | Student Lounge | 192.192.0.0 | 192.223.255.255 | 192.192.0.1 - 192.223.255.254 | /11 255.224.0.0 |
| 7 | Video Conferencing Room | 192.224.0.0 | 192.255.255.255 | 192.224.0.1 - 192.255.255.254 | /11 255.224.0.0 |

Network distribution for workstation in different work areas.

Ground Floor:

1. General Lab 1

| Device | IP Address |
|---|---|
| PC1 | 192.32.0.1 |
| PC2 | 192.32.0.2 |
| PC3 | 192.32.0.3 |

| | |
|---|---|
| PC4 | 192.32.0.4 |
| PC5 | 192.32.0.5 |
| PC6 | 192.32.0.6 |
| PC7 | 192.32.0.7 |
| PC8 | 192.32.0.8 |
| PC9 | 192.32.0.9 |
| PC10 | 192.32.0.10 |
| PC11 | 192.32.0.11 |
| PC12 | 192.32.0.12 |
| PC13 | 192.32.0.13 |
| PC14 | 192.32.0.14 |
| PC15 | 192.32.0.15 |
| PC16 | 192.32.0.16 |
| PC17 | 192.32.0.17 |
| PC18 | 192.32.0.18 |
| PC19 | 192.32.0.19 |
| PC20 | 192.32.0.20 |
| PC21 | 192.32.0.21 |
| PC22 | 192.32.0.22 |
| PC23 | 192.32.0.23 |
| PC24 | 192.32.0.24 |
| PC25 | 192.32.0.25 |
| PC26 | 192.32.0.26 |
| PC27 | 192.32.0.27 |
| PC28 | 192.32.0.28 |
| PC29 | 192.32.0.29 |
| PC30 | 192.32.0.30 |
| PC31 | 192.32.0.31 |
| Projector 1 | 192.32.0.32 |
| WAP 1 | 192.32.0.33 |
| WAP 2 | 192.32.0.34 |

2. General Lab 2

| Device | IP Address |
|---|---|
| PC32 | 192.64.0.1 |
| PC33 | 192.64.0.2 |
| PC34 | 192.64.0.3 |
| PC35 | 192.64.0.4 |
| PC36 | 192.64.0.5 |
| PC37 | 192.64.0.6 |
| PC38 | 192.64.0.7 |
| PC36 | 192.64.0.8 |
| PC40 | 192.64.0.9 |
| PC41 | 192.64.0.10 |

| | |
|---|---|
| PC42 | 192.64.0.11 |
| PC43 | 192.64.0.12 |
| PC44 | 192.64.0.13 |
| PC45 | 192.64.0.14 |
| PC46 | 192.64.0.15 |
| PC47 | 192.64.0.16 |
| PC48 | 192.64.0.17 |
| PC49 | 192.64.0.18 |
| PC50 | 192.64.0.19 |
| PC51 | 192.64.0.20 |
| PC52 | 192.64.0.21 |
| PC53 | 192.64.0.22 |
| PC54 | 192.64.0.23 |
| PC55 | 192.64.0.24 |
| PC56 | 192.64.0.25 |
| PC57 | 192.64.0.26 |
| PC58 | 192.64.0.27 |
| PC59 | 192.64.0.28 |
| PC60 | 192.64.0.29 |
| PC61 | 192.64.0.30 |
| PC62 | 192.64.0.31 |
| Projector 2 | 192.64.0.32 |
| WAP 3 | 192.64.0.33 |

3. Cisco Networking Lab

| Device | IP Address |
|---|---|
| PC63 | 192.96.0.1 |
| PC64 | 192.96.0.2 |
| PC65 | 192.96.0.3 |
| PC66 | 192.96.0.4 |
| PC67 | 192.96.0.5 |
| PC68 | 192.96.0.6 |
| PC69 | 192.96.0.7 |
| PC70 | 192.96.0.8 |
| PC71 | 192.96.0.9 |
| PC72 | 192.96.0.10 |
| PC73 | 192.96.0.11 |
| PC74 | 192.96.0.12 |
| PC75 | 192.96.0.13 |
| PC76 | 192.96.0.14 |
| PC77 | 192.96.0.15 |
| PC78 | 192.96.0.16 |
| PC79 | 192.96.0.17 |
| PC80 | 192.96.0.18 |
| PC81 | 192.96.0.19 |
| PC82 | 192.96.0.20 |

| | |
|---|---|
| PC83 | 192.96.0.21 |
| PC84 | 192.96.0.22 |
| PC85 | 192.96.0.23 |
| PC86 | 192.96.0.24 |
| PC87 | 192.96.0.25 |
| PC88 | 192.96.0.26 |
| PC89 | 192.96.0.27 |
| PC90 | 192.96.0.28 |
| PC91 | 192.96.0.29 |
| PC92 | 192.96.0.30 |
| Projector 3 | 192.96.0.31 |
| WAP 4 | 192.96.0.32 |

4. Embedded Lab

| Device | IP Address |
|---|---|
| PC93 | 192.128.0.1 |
| PC94 | 192.128.0.2 |
| PC95 | 192.128.0.3 |
| PC96 | 192.128.0.4 |
| PC97 | 192.128.0.5 |
| PC98 | 192.128.0.6 |
| PC99 | 192.128.0.7 |
| PC100 | 192.128.0.8 |
| PC101 | 192.128.0.9 |
| PC102 | 192.128.0.10 |
| PC103 | 192.128.0.11 |
| PC104 | 192.128.0.12 |
| PC105 | 192.128.0.13 |
| PC106 | 192.128.0.14 |
| PC107 | 192.128.0.15 |
| PC108 | 192.128.0.16 |
| PC109 | 192.128.0.17 |
| PC110 | 192.128.0.18 |
| PC111 | 192.128.0.19 |
| PC112 | 192.128.0.20 |
| PC113 | 192.128.0.21 |
| PC114 | 192.128.0.22 |
| PC115 | 192.128.0.23 |
| PC116 | 192.128.0.24 |
| PC117 | 192.128.0.25 |
| PC118 | 192.128.0.26 |
| PC119 | 192.128.0.27 |
| PC120 | 192.128.0.28 |
| PC121 | 192.128.0.29 |
| PC122 | 192.128.0.30 |
| PC123 | 192.128.0.31 |

| | |
|---|---|
| Projector 4 | 192.128.0.32 |
| WAP 5 | 192.128.0.33 |

Ground floor with label:

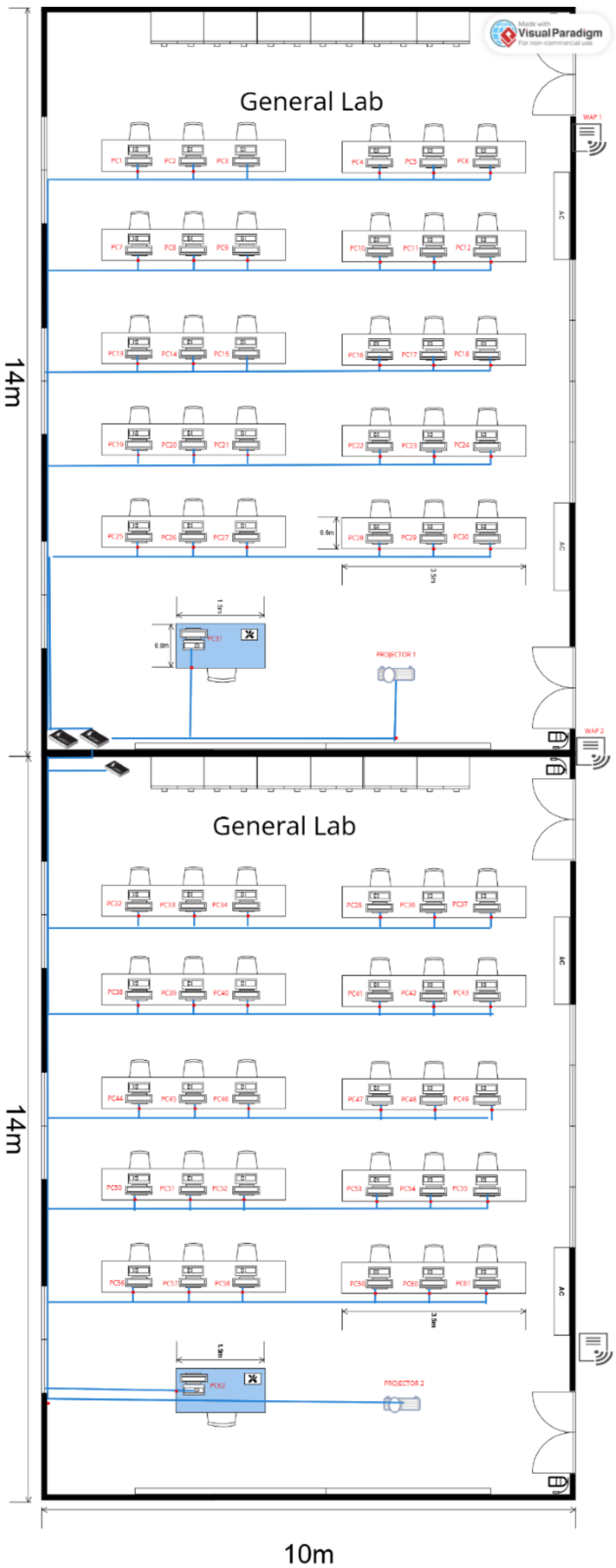



Figure 5.2.1 General Lab 1 & 2 with label

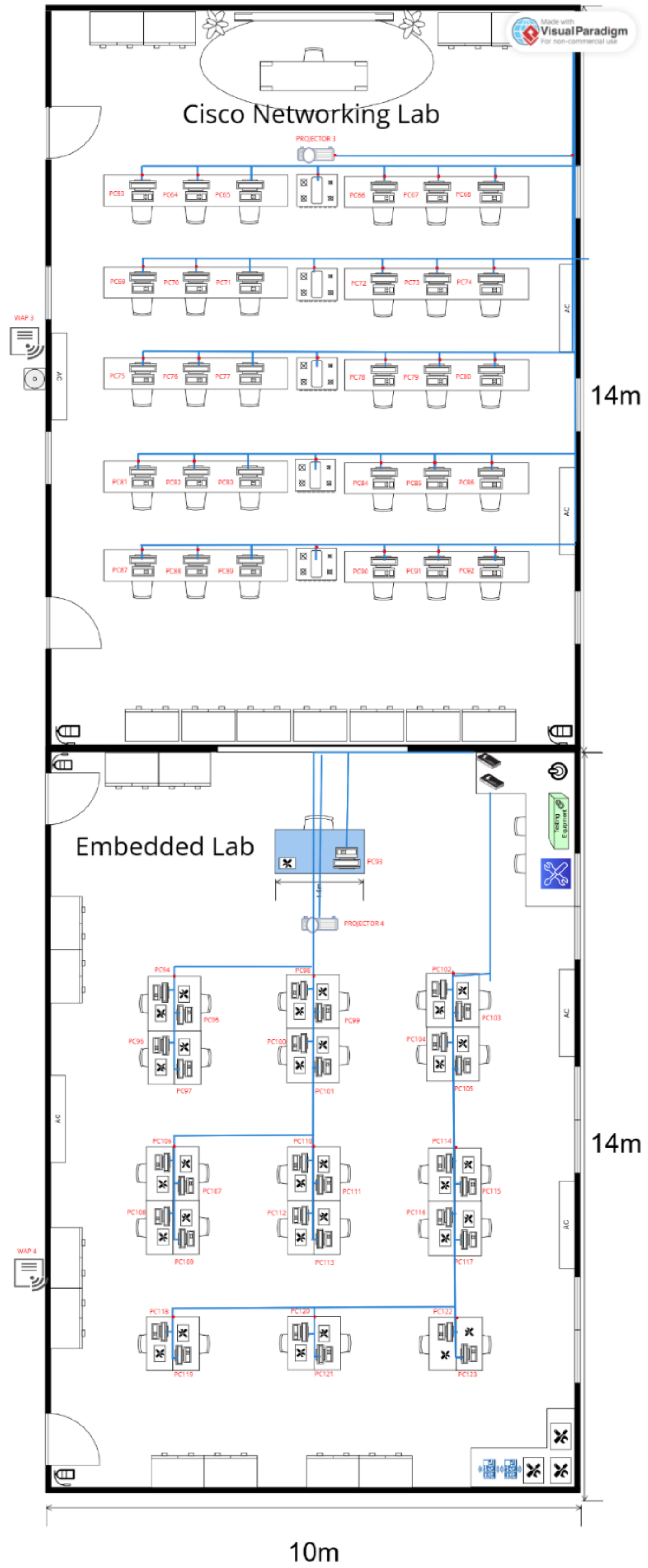



Figure 5.2.2 Cisco Networking Lab and Embeded Lab with label

First Floor:

5. Hybrid Classroom

| Device | IP Address |
|---|---|
| PC124 | 192.160.0.1 |
| PC125 | 192.160.0.2 |
| PC126 | 192.160.0.3 |
| PC127 | 192.160.0.4 |
| PC128 | 192.160.0.5 |
| PC129 | 192.160.0.6 |
| PC130 | 192.160.0.7 |
| PC131 | 192.160.0.8 |
| PC132 | 192.160.0.9 |
| PC133 | 192.160.0.10 |
| PC134 | 192.160.0.11 |
| PC135 | 192.160.0.12 |
| PC136 | 192.160.0.13 |
| PC137 | 192.160.0.14 |
| PC138 | 192.160.0.15 |
| PC139 | 192.160.0.16 |
| PC140 | 192.160.0.17 |
| PC141 | 192.160.0.18 |
| PC142 | 192.160.0.19 |
| PC143 | 192.160.0.20 |
| PC144 | 192.160.0.21 |
| PC145 | 192.160.0.22 |
| PC146 | 192.160.0.23 |
| PC147 | 192.160.0.24 |
| PC148 | 192.160.0.25 |
| PC149 | 192.160.0.26 |
| PC150 | 192.160.0.27 |
| PC151 | 192.160.0.28 |
| PC152 | 192.160.0.29 |
| PC153 | 192.160.0.30 |
| PC154 | 192.160.0.31 |
| Projector 5 | 192.160.0.32 |
| WAP 6 | 192.160.0.33 |

6. Student Lounge

| Device | IP Address |
|---|---|
| WAP 7 | 192.192.0.1 |

7. Video Conferencing Room

| Device | IP Address |
|---|---|

| | |
|---|---|
| IoT 1 | 192.224.0.1 |
| WAP 8 | 192.224.0.2 |

8. Staff Office

| Device | IP Address |
|---|---|
| PC155 | 192.0.0.1 |
| PC156 | 192.0.0.2 |
| PC157 | 192.0.0.3 |
| PC158 | 192.0.0.4 |
| PC159 | 192.0.0.5 |
| PC160 | 192.0.0.6 |
| PC161 | 192.0.0.7 |
| PC162 | 192.0.0.8 |
| PC163 | 192.0.0.9 |
| PC164 | 192.0.0.10 |
| WAP 9 | 192.0.0.11 |

9. Server Room

| Device | IP Address |
|---|---|
| Router 1 | 192.0.0.12 |
| Router 2 | 192.0.0.13 |
| Router 3 | 192.0.0.14 |
| Router 4 | 192.0.0.15 |
| Server 1 | 192.0.0.16 |
| Server 2 | 192.0.0.17 |
| Server 3 | 192.0.0.18 |
| Server 4 | 192.0.0.19 |
| WAP 10 | 192.0.0.20 |

First floor with label:

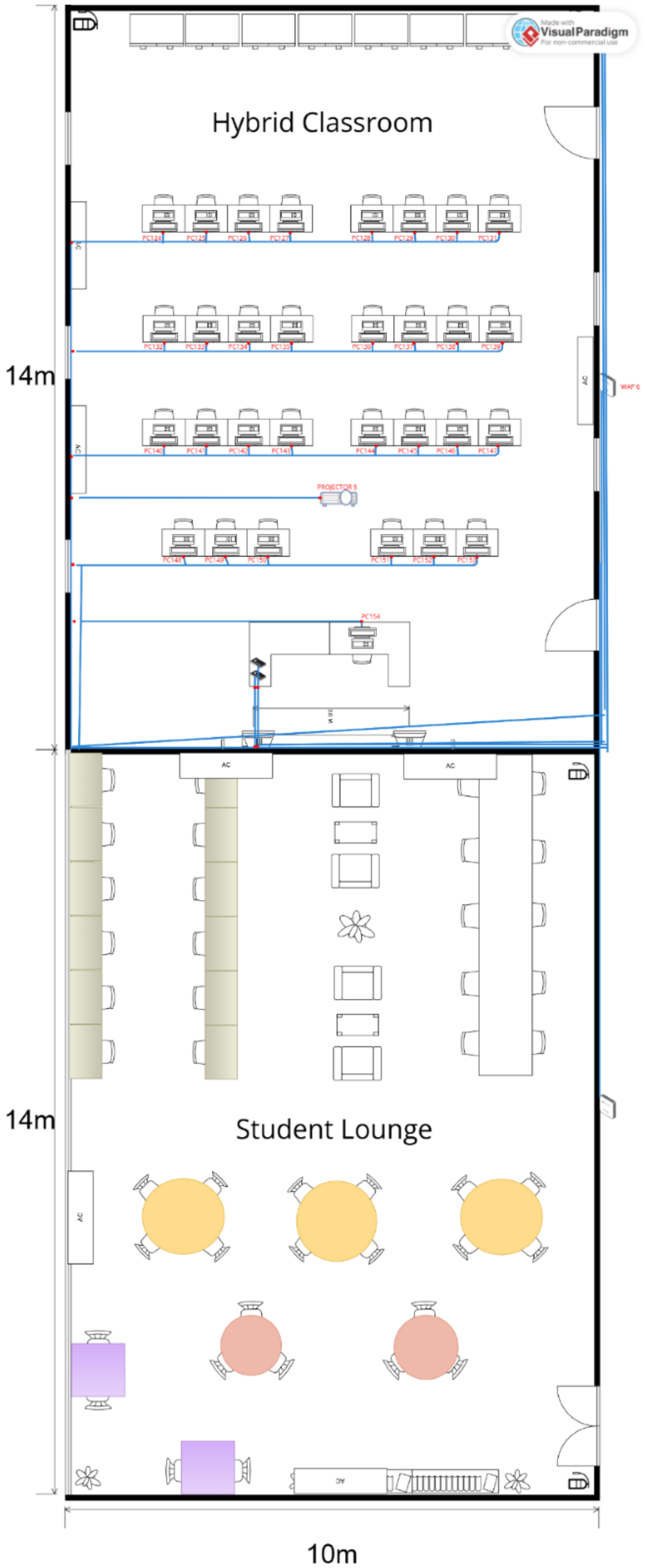



Figure 5.2.3 Hybrid Classroom and Student Lounge with label

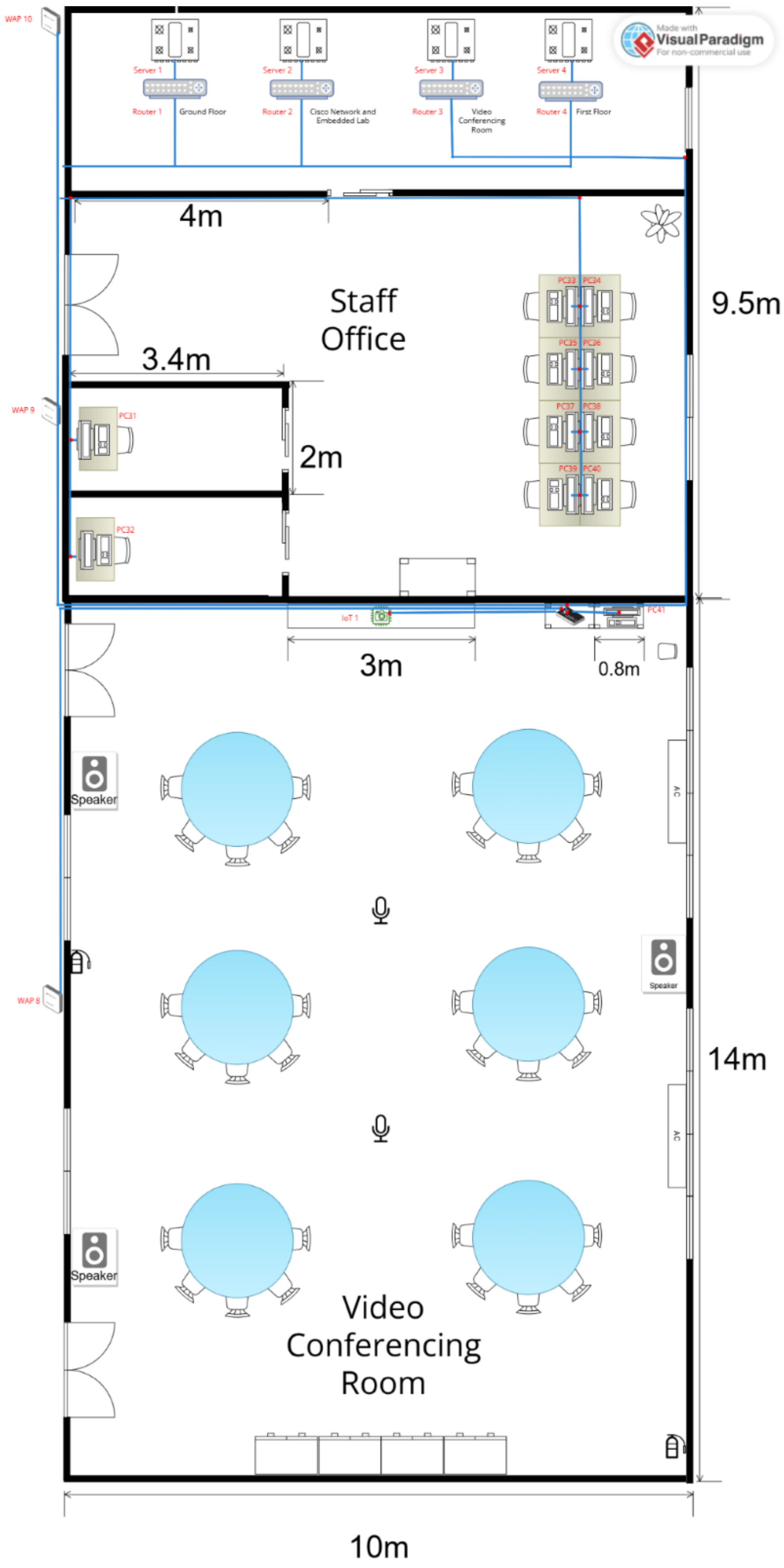



Figure 5.2.4 Server Room, Staff Office and Video Conferencing Room with label

### 4.5.1 Task 5 Reflection

Completing Task 5 was a collaborative effort that showcased the strengths and teamwork of our group. Each member contributed unique skills, whether it was in calculating subnetting, designing network layouts or managing documentation. We divided the responsibilities effectively. We separated tasks into 2 parts which is calculate subnetting and divide all IP address into subnets and distribute it into each device. This division allowed us to work efficiently and meet the deadlines while learning from each other's expertise.

One of the major lessons we learned during this task was the importance of proper subnetting in network design. By understanding how to allocate IP addresses efficiently, we ensured that different departments and work areas within the network had sufficient resources. We also realized how critical it is to plan for future scalability and network security while designing the IP structure. This task deepened out understanding of IP addressing, subnet masks and the practical application of these concepts in a real-world scenario.

Additionally, this task emphasized the value of communication and collaboration within a team. Regular discussions and meetings help us align our ideas and troubleshoot any issues that arose. We learned how to delegate tasks based on individual strengths and how to integrate everyone's work into a cohesive final submission. The process improved our problem-solving skills and taught us to approach complex technical tasks methodically.

Overall, Task 5 was an invaluable learning experience that not only strengthened our technical knowledge but also enhances our teamwork abilities. We now have a better grasp of network design principles, and the practical steps involved in creating efficient and secure networks. The task also highlighted the importance of thorough planning and collaboration, lessons that will benefit us in future projects and professional settings.

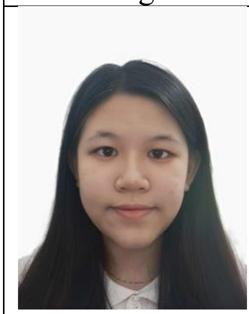
# 5.0 Conclusion

# 6.0 Additional Considerations for Informed Decision-Making

**Cost Optimization Through Bulk Purchasing and Vendor Negotiations**

A highly effective way to optimize costs is through bulk purchasing and vendor negotiations, which can significantly reduce expenses while adding value. Many suppliers offer discounts for large orders, making it more cost-efficient to buy networking equipment in bundles rather than individually. By negotiating comprehensive package deal that includes routers, access points, switches, and cabling from a single supplier, organizations can achieve a 15% to 20% reduction in total costs. Additionally, some vendors provide free onsite support, extended warranties, and maintenance plans, which help minimize future repair and replacement expenses. These strategic negotiations not only lower initial investment costs but also contribute to long-term network stability and efficiency, ensuring a more sustainable and cost-effective infrastructure.

# 7.0 Team Member and Responsibilities

| Team Member | Information and Responsibilities |
|---|---|
| <br>Poh Lok Yee | Matric Number: A23CS0262 |
| <br>Chau Ying Jia | Matric Number: A23CS0213 |
| <br>Sabrina Heng Wei Qi | Matric Number: A23CS0265 |
| <br>Woo Cheng Shuan | Matric Number: A23CS0283 |

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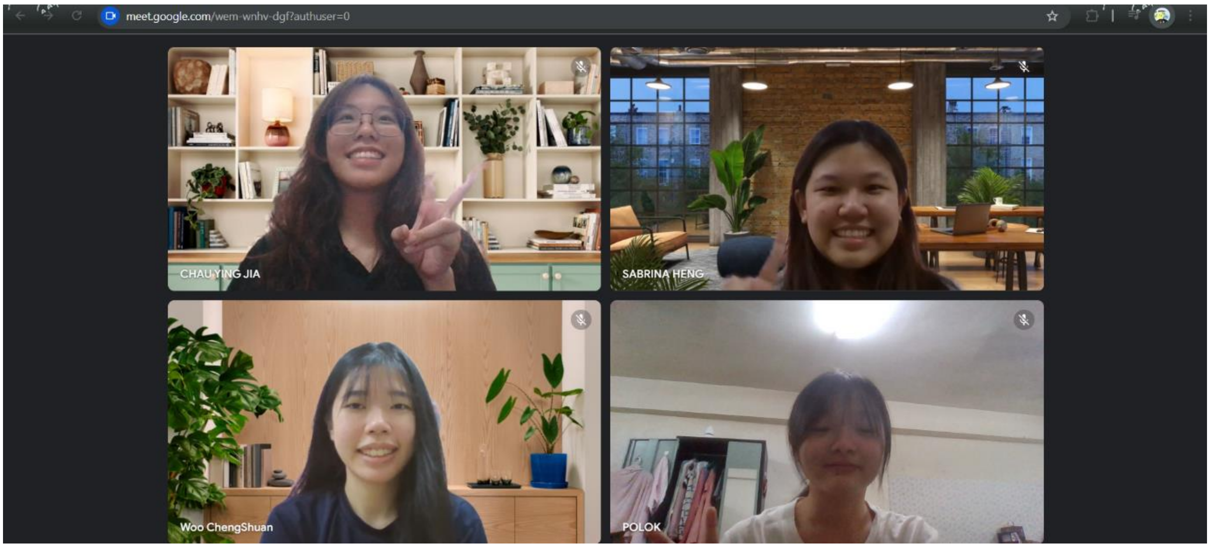
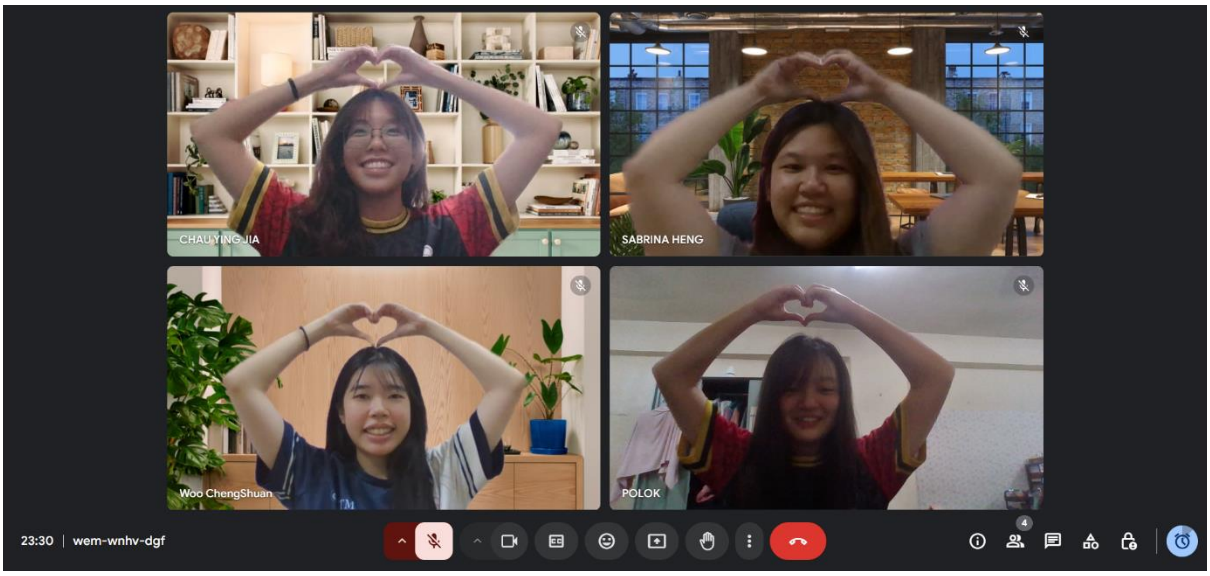
# 9.0 Appendices







# MEETING MINUTES #1

| | | |
|---|---|---|
| **DATE/TIME** | *4 Nov 2024 3pm* | |
| **LOCATION** | Arked Angkasa | |
| **AGENDA** | 1. *Discuss about all the task from task 1*<br>2. *Tool for designing floor plan*<br>3. *Floor plan arrangement*<br>4. *Task distribution for each person* | |
| **Meeting MC** | *Poh Lok Yee* | |
| **ATTENDANCE** | | |
| **NAME** | **TIME** | **REASON FOR ABSENCE** |
| Poh Lok Yee | 1500 | - |
| Woo Cheng Shuan | 1500 | - |
| Chau Ying Jia | 1500 | - |
| Sabrina Heng Wei Qi | 1520 | - |

**MINUTES**

| **NO.** | **ITEM DISCUSSED** | **IDEAS/SUGGESTIONS AND PERSON GIVING IT** | **PERSON IN CHARGE** |
|---|---|---|---|
| 1 | Suggest group name | *All members suggested 4fabs as group name* | All members |
| 2 | Shirt discussion on Task 1 | *All members read and understand the task given* | All members |
| 3 | Software to use | *Ying Jia suggested to use visual paradigm to draw floor plan* | *Ying Jia* |
| 4 | Building design | *Ying Jia and Cheng Shuan suggested building design in rectangle shape*<br><br>*Lok Yee suggested building design in octagon shape* | Ying Jia and Cheng Shuan |
| 5 | Decision on building design | *Sabrina, Ying Jia and Cheng Shuan drew the draft for rectangle building design* | All members |

| | | | |
|---|---|---|---|
| | | *Lok Yee drew the draft for octagon building design*<br><br>*All members discuss on the advantages and disadvantages between the designs and make final decision on building design* | |
| 6 | Floor plan design | *All members discuss floor plan design*<br><br>*Lok Yee suggested adding lift and accessible toilet for disability friendly*<br><br>*Cheng Shuan suggested all 4 labs at the lower ground*<br><br>*Sabrina suggested the size for Video conferencing room and hybrid classroom*<br><br>*Ying Jia drew the draft of floor plan design* | All members |
| 7 | Task distribution | *Lok Yee distributes the tasks for each member by using spin tools*<br><br>*Outcomes:*<br><br>*Lok Yee is responsible for designing the student lounge and first floor, Ying Jia is responsible for designing the second floor and Embedded lab, Sabrina is responsible for designing Cisco Network lab and hybrid classroom while Cheng Shuan is responsible for designing the general lab and Video Conferencing room* | Lok Yee |
| 8 | Next Meeting | *8 November – All given tasks should be 90% completed* | All members |
| 9 | Meeting ended | *1715* | All members |

# MEETING MINUTES #2

| | | |
|---|---|---|
| **DATE/TIME** | *8 Nov 2024 2pm* | |
| **LOCATION** | KTDI MA7 | |
| **AGENDA** | 1. *Finalize the floor plan*<br>2. *Learn packet tracer* | |
| **Meeting MC** | *Chau Ying Jia* | |
| **ATTENDANCE** | | |
| **NAME** | **TIME** | **REASON FOR ABSENCE** |
| Poh Lok Yee | 1400 | - |
| Woo Cheng Shuan | 1400 | - |
| Chau Ying Jia | 1400 | - |
| Sabrina Heng Wei Qi | 1400 | - |

**MINUTES**

| **NO.** | **ITEM DISCUSSED** | **IDEAS/SUGGESTIONS AND PERSON GIVING IT** | **PERSON IN CHARGE** |
|---|---|---|---|
| 1 | Sharing session on current progress | *All members shared own progresses* | All members |
| 2 | Comment and give suggestion for enhancement | *Lok Yee reminded all labs must have 30 workstations*<br>*Cheng Shuan suggested all the same elements using same scale*<br>*Sabrina suggested the size of stairs and lift*<br>*Ying Jia suggested both floor toilets using same design* | All members |
| 3 | Design and edit each room and floor | *All members edit the rooms following the suggestion and reminder from other group members* | *All members* |
| 4 | Combination of each lab and room on both | *All members put each of their designs inside the ground* | All members |

| | | | |
|---|---|---|---|
| | floors and finalize the floor plan design | *floor and first floor and finalize the design* | |
| 5 | Next Meeting | *14 November – All members should download and learn how to use packet tracer* | All members |
| 6 | Meeting Ended | *1800* | All members |

# MEETING MINUTES #3

| | | |
|---|---|---|
| **DATE/TIME** | *14 Nov 2024 3pm* | |
| **LOCATION** | KTDI MA7 | |
| **AGENDA** | 1. *Use Packet Tracer to enhance the floor plan*<br>2. *Finalize the floor plan* | |
| **Meeting MC** | *Sabrina Heng Wei Qi* | |
| **ATTENDANCE** | | |
| **NAME** | **TIME** | **REASON FOR ABSENCE** |
| Poh Lok Yee | 1500 | - |
| Woo Cheng Shuan | 1500 | - |
| Chau Ying Jia | 1500 | - |
| Sabrina Heng Wei Qi | 1500 | - |

| **MINUTES** | | | |
|---|---|---|---|
| **NO.** | **ITEM DISCUSSED** | **IDEAS/SUGGESTIONS AND PERSON GIVING IT** | **PERSON IN CHARGE** |
| 1 | Discuss function packet tracer | *All members shared their own understanding on packet tracer* | All members |
| 2 | Use Packet Tracer to enhance the floor plan design | *Each member use Packet Tracer to design their assigned room.* | All members |
| 3 | Finalize the floor plan | *Finalize the floor plan* | All members |
| 4 | Meeting Ended | *1845* | All members |

**MEETING MINUTES #4**

| | | | |
|---|---|---|---|
| **DATE/TIME** | *22 Nov 2024 10am* | | |
| **LOCATION** | Online Webex Meeting | | |
| **AGENDA** | 1. *Discuss about all the task from task 2*<br>2. *Suggest possible questions for task 2*<br>3. *Select the most related questions from all suggestion*<br>4. *Task distribution for each person* | | |
| **Meeting MC** | *Woo Cheng Shuan* | | |
| **ATTENDANCE** | | | |
| **NAME** | **TIME** | | **REASON FOR ABSENCE** |
| Poh Lok Yee | 1000 | | - |
| Woo Cheng Shuan | 1000 | | - |
| Chau Ying Jia | 1000 | | - |
| Sabrina Heng Wei Qi | 1000 | | - |
| **MINUTES** | | | |

| **NO.** | **ITEM DISCUSSED** | **IDEAS/SUGGESTIONS AND PERSON GIVING IT** | **PERSON IN CHARGE** |
|---|---|---|---|
| 1 | Discussion for task 2 | *All members read and understand requirements from task2.* | All members |
| 2 | Suggest possible questions | *Each member contributes 5-6 questions regarding the case study.* | All members |
| 3 | Select the most related questions | *Sabrina lists out the questions from each member in the Word document to easier members to read in overall.*<br><br>*All members read and understand the questions provided by other members.* | All members |

| | | | |
|---|---|---|---|
| | | *All members discussed the questions one by one and selected 20 most related questions that can be used in task 2.* | |
| 4 | Task distribution | *Lok Yee distributed the task to all members equally.*<br><br>*Each member gets 5 questions.*<br><br>*All members had to do research to find a solution for those questions with listing out the references.* | Lok Yee |
| 5 | Next Meeting | *26 November – All given tasks should be 95% completed* | All members |
| 6 | Meeting ended | *1300* | All members |

**MEETING MINUTES #5**

| | | | |
|---|---|---|---|
| **DATE/TIME** | *26 Nov 2024 10am* | | |
| **LOCATION** | Online Webex Meeting | | |
| **AGENDA** | 1. *Review the answers for each question*<br>2. *Finalize the most suitable questions to be used*<br>3. *Discuss on the feasibility assessment* | | |
| **Meeting MC** | *Poh Lok Yee* | | |
| **ATTENDANCE** | | | |
| **NAME** | **TIME** | | **REASON FOR ABSENCE** |
| Poh Lok Yee | 1000 | | - |
| Woo Cheng Shuan | 1000 | | - |
| Chau Ying Jia | 1000 | | - |
| Sabrina Heng Wei Qi | 1000 | | - |
| **MINUTES** | | | |
| **NO.** | **ITEM DISCUSSED** | **IDEAS/SUGGESTIONS AND PERSON GIVING IT** | **PERSON IN CHARGE** |
| 1 | Explain answer for each question | *All members explained their responsive questions and answers one by one.* | All members |
| 2 | Review the answer | *All members review the questions and answer carefully* | All members |
| 3 | Finalize the most related questions and answers | *Sabrina suggested dropping the questions with answers that are not from any reference.*<br><br>*Ying Jia suggested dropping the questions that are quite related.* | All members |

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|---|---|---|---|
| | | *Cheng Shuan said that some answers are confusing.*<br><br>*Lok Yee highlighted the question that can be used.*<br><br>*Lok Yee finalizes 15 questions and answers.*<br><br>*All members agree with the finalized list.* | |
| 4 | Discuss on the feasibility assessment | *Lok Yee explained the feasibility assessment to all members.*<br><br>*Ying Jia suggested covering several aspects in this assessment.* | Lok Yee and Ying Jia |
| 5 | Meeting ended | *At 1230, the meeting ended after all discussions were done.* | All members |

# MEETING MINUTES #6

| | | |
|---|---|---|
| **DATE/TIME** | *28 Dec 2024 4pm* | |
| **LOCATION** | MA7 | |
| **AGENDA** | 1. *Discuss about all the task from task 3*<br>2. *Suggest devices needed in task 3*<br>3. *Research about the price of all devices*<br>4. *Select the most suitable devices from all the choices*<br>5. *Task distribution for each person* | |
| **Meeting MC** | *Sabrina Heng Wei Qi* | |
| **ATTENDANCE** | | |
| **NAME** | **TIME** | **REASON FOR ABSENCE** |
| Poh Lok Yee | 1600 | - |
| Woo Cheng Shuan | 1600 | - |
| Chau Ying Jia | 1600 | - |
| Sabrina Heng Wei Qi | 1600 | - |

**MINUTES**

| NO. | ITEM DISCUSSED | IDEAS/SUGGESTIONS AND PERSON GIVING IT | PERSON IN CHARGE |
|---|---|---|---|
| 1 | Discussion for task 3 | *All members read and understand requirements from task3.* | All members |
| 2 | Suggest devices needed | *Sabrina and Ying Jia search and list out the suggested devices needed in ground floor of the building.*<br><br>*Lok Yee and Cheng Shuan search and list out the suggested devices needed* | All members |

| | | | |
|---|---|---|---|
| | | *in first floor of the building.*<br><br>*Ying Jia creates a table for listing out all the devices to easier members to read in overall.* | |
| 3 | Research about the prices of all devices | *Sabrina and Ying Jia search for the price of devices needed in ground floor of the building.*<br><br>*Lok Yee and Cheng Shuan search for the price of devices needed in first floor of the building.* | All members |
| 4 | Select most suitable device | *All members discuss about the Pros and cons of each device and compare the price of each device.*<br><br>*All members vote for the most suitable device to use in the building.* | All members |
| 5 | Task distribution | *Sabrina distributed the task to all members equally.*<br><br>*All members had to do research on each device, list out the device quantity used, price and others* | Sabrina Heng Wei Qi |
| 6 | Next Meeting | *6 January – All given tasks should be 95% completed* | All members |
| 7 | Meeting ended | *1900* | All members |

## MEETING MINUTES #7

| | |
|---|---|
| **DATE/TIME** | *12 Jan 2025 7pm* |

| | | | |
|---|---|---|---|
| **LOCATION** | MA7 | | |
| **AGENDA** | 1. *Discuss about all the task from task 4*<br>2. Discuss about the network distribution in our floor plan.<br>3. *Discuss about the details in designing the network routes*<br>4. *Task distribution for members* | | |
| **Meeting MC** | *Poh Lok Yee* | | |
| **ATTENDANCE** | | | |
| **NAME** | **TIME** | **REASON FOR ABSENCE** | |
| Poh Lok Yee | 1900 | - | |
| Woo Cheng Shuan | 1900 | - | |
| Chau Ying Jia | 1900 | - | |
| Sabrina Heng Wei Qi | 1900 | - | |

| | | | |
|---|---|---|---|
| **MINUTES** | | | |
| **NO.** | **ITEM DISCUSSED** | **IDEAS/SUGGESTIONS AND PERSON GIVING IT** | **PERSON IN CHARGE** |
| 1 | Discussion for task 4 | *Poh Lok Yee brief us the information and rubrics in task 4 All members read and understand requirements from task3.* | Poh Lok Yee |
| 2 | Suggestion for network distribution | *Lok Yee and Ying Jia provide ideas in designing the network distribution in the floor plan.*<br><br>*Sabrina and Cheng Shuan suggest the necessary details that need to be standardized.* | All members |
| 3 | Task Distribution | *Lok Yee distributed tasks to all members.*<br><br>*Cheng Shuan and Sabrina was assigned to have an idea and plan the overall network of building* | All members |

| | | | |
|---|---|---|---|
| | | *Ying Jia was assigned to plan the details in the distribution of network for the first floor*<br><br>*Lok Yee will be planning for the details in ground floor network distribution*<br><br>*All members will be calculating the usage of networking cables and stuff together* | |
| 4 | Meeting Ended | *Meeting ended after all discussions have done* | All members |

# MEETING MINUTES #8

| | | |
|---|---|---|
| **DATE/TIME** | *19 Jan 2025 8pm* | |
| **LOCATION** | Google meet (online) | |
| **AGENDA** | *1. Review the current progress for each group member.*<br>*2. Finalize network distribution diagram*<br>*3. Check the calculation of the cable length needed* | |
| **Meeting MC** | *Woo Cheng Shuan* | |
| **ATTENDANCE** | | |
| **NAME** | **TIME** | **REASON FOR ABSENCE** |
| Poh Lok Yee | 2000 | - |
| Woo Cheng Shuan | 2000 | - |
| Chau Ying Jia | 2000 | - |
| Sabrina Heng Wei Qi | 2000 | - |

| **MINUTES** | | | |
|---|---|---|---|
| **NO.** | **ITEM DISCUSSED** | **IDEAS/SUGGESTIONS AND PERSON GIVING IT** | **PERSON IN CHARGE** |
| 1 | Review the current<br>progress for each group member | *Each group member present their parts* | All members |
| 2 | Finalize network distribution diagram | *All members give suggestion and discuss on what should be implemented and change before finalising the network distribution* | All members |
| 3 | Calculation of the cable length needed | *We calculate the total length of wire together and give* | All members |

| | | | |
|---|---|---|---|
| | | *suggestion and discuss about the calculation together* | |
| 4 | Meeting Ended | *Meeting ended after all discussions have done* | All members |

# Meeting Minutes #9

<table>
<tr><td colspan="2">DATE/TIME</td><td colspan="2">22 Jan 2025 8pm</td></tr>
<tr><td colspan="2">LOCATION</td><td colspan="2">Google Meeting</td></tr>
<tr><td colspan="2">AGENDA</td><td colspan="2">1. Discuss about all the task from task 5<br>2. Discuss about the network division<br>3. Complete task 5 together</td></tr>
<tr><td colspan="2">Meeting MC</td><td colspan="2">Chau Ying Jia</td></tr>
<tr><td colspan="4">ATTENDANCE</td></tr>
<tr><td colspan="2">NAME</td><td>TIME</td><td>REASON FOR ABSENCE</td></tr>
<tr><td colspan="2">Poh Lok Yee</td><td>2000</td><td>-</td></tr>
<tr><td colspan="2">Woo Cheng Shuan</td><td>2000</td><td>-</td></tr>
<tr><td colspan="2">Chau Ying Jia</td><td>2000</td><td>-</td></tr>
<tr><td colspan="2">Sabrina Heng Wei Qi</td><td>2000</td><td>-</td></tr>
<tr><td colspan="4">MINUTES</td></tr>
<tr><td>NO.</td><td>ITEM DISCUSSED</td><td>IDEAS/SUGGESTIONS AND PERSON GIVING IT</td><td>PERSON IN CHARGE</td></tr>
<tr><td>1</td><td>Discussion for task 5</td><td>Chau Ying Jia brief us the information and rubrics in task 5<br>All members read and understand requirements from task 5.</td><td>Chau Ying Jia</td></tr>
<tr><td>2</td><td>Suggestion for network division</td><td>All members suggest their prefer subnetwork division and decide the one best suit our design</td><td>All members</td></tr>
<tr><td>3</td><td>IP Addressing Scheme Overview</td><td>All members examine how IP addresses are currently assigned to each work area.</td><td>All members</td></tr>
<tr><td>4</td><td>Task Distribution</td><td>Ying Jia distributed the tasks for all members.</td><td>All members</td></tr>
<tr><td>5</td><td>Proceed Tasks 5 together</td><td>All members work on Words together to complete task 5.</td><td>All members</td></tr>
</table>

| | | | |
|---|---|---|---|
| 6 | Meeting Ended | *Meeting ended after all discussions have done* at 2300 | All members |